



Design and manage analytics solutions using Power BI

Day 1

1

Get started with Microsoft data analytics

© Copyright Microsoft Corporation. All rights reserved.

2

From data to business insights with Power BI



© Copyright Microsoft Corporation. All rights reserved.

3

Roles in Data



Business Analyst



Data Analyst



Data Engineer



Analytics Engineer

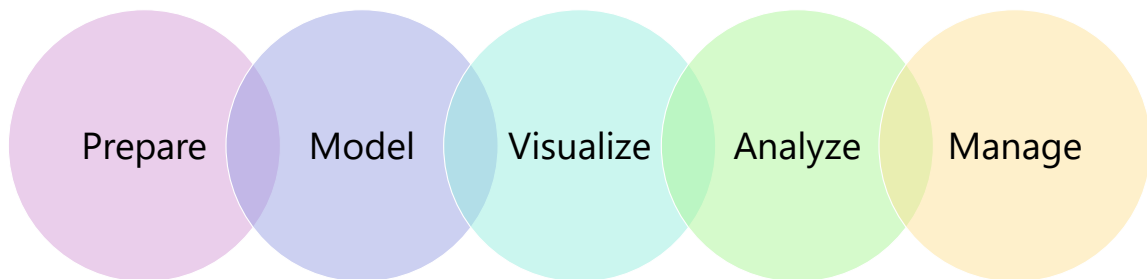


Data Scientist

© Copyright Microsoft Corporation. All rights reserved.

4

Tasks of a Data Analyst



© Copyright Microsoft Corporation. All rights reserved.

5

Different types of analysis

Data analysis is another form of storytelling with five categories:

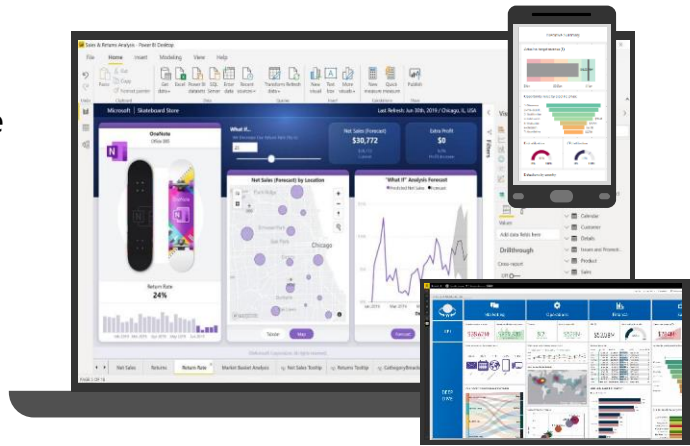
- **Descriptive:** Summarize past data
- **Diagnostic:** Explain past data
- **Predictive:** Forecast future data
- **Prescriptive:** Optimize future data
- **Artificial Intelligence:** Learn from data

© Copyright Microsoft Corporation. All rights reserved.

6

Introduction to Power BI environment

- Create with Power BI Desktop
- Distribute with Power BI service
- Access with Power BI Mobile



© Copyright Microsoft Corporation. All rights reserved.

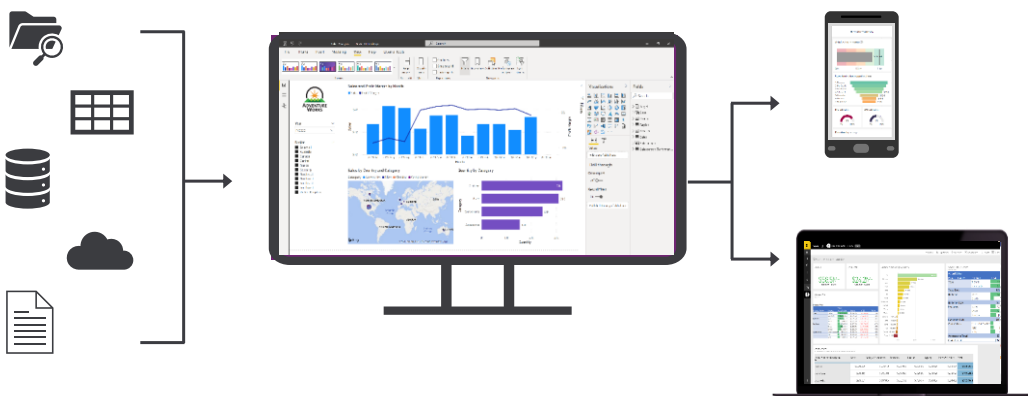
7

The flow of Power BI

Data sources

Power BI Desktop

Power BI service & Mobile



© Copyright Microsoft Corporation. All rights reserved.

8

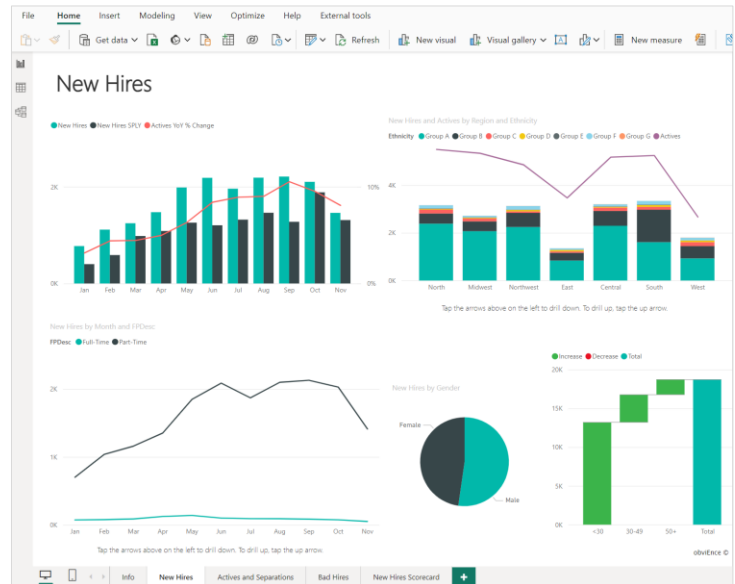
The building blocks of Power BI

Semantic models

- Connected data, transformations, relationships, and calculations.

Visualizations

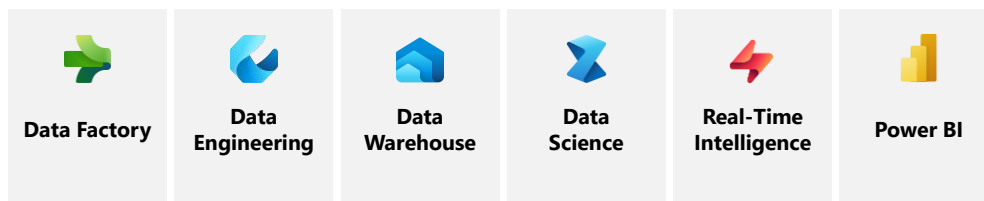
- Interactive representation of data through various charts and graphics.



© Copyright Microsoft Corporation. All rights reserved.

9

Microsoft Fabric



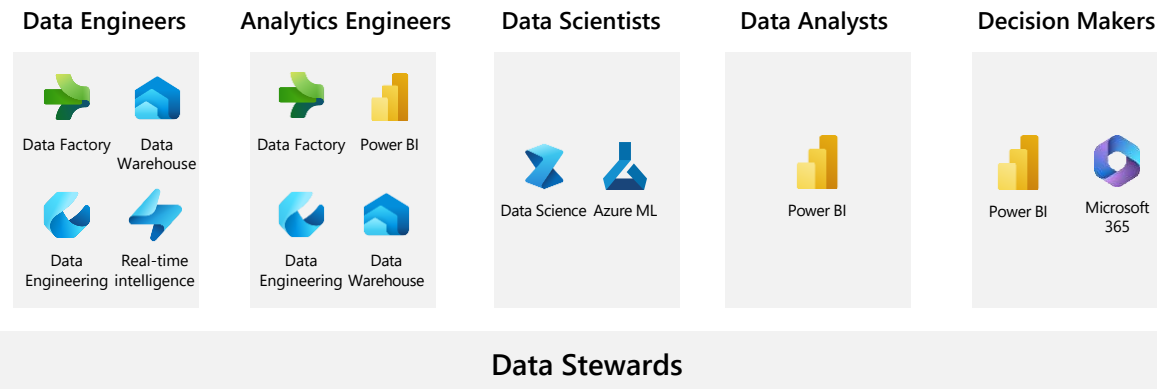
- **Data Factory:** Data integration combining Power Query with the scale of Azure Data Factory to move and transform data.
- **Data Engineering:** Data engineering with a Spark platform for data transformation at scale.
- **Data Warehouse:** Data warehousing with industry-leading SQL performance and scale to support data use.
- **Data Science:** Data science with Azure Machine Learning and Spark for model training and execution tracking in a scalable environment.
- **Real-Time Intelligence:** Real-time Intelligence to query and analyze large volumes of data in real-time.
- **Power BI:** Business intelligence for translating data to decisions.

© Copyright Microsoft Corporation. All rights reserved.

10

Data teams and Fabric

Fabric's unified management and governance make it easier for data professionals to work together.



© Copyright Microsoft Corporation. All rights reserved.

11

Further reading

Discover data analysis

learn.microsoft.com/training/modules/data-analytics-microsoft/

Get building with Microsoft Power BI

<https://learn.microsoft.com/training/modules/get-started-with-power-bi/>

Introduction to end-to-end analytics using Microsoft Fabric

<https://aka.ms/fabric-intro>



© Copyright Microsoft Corporation. All rights reserved.

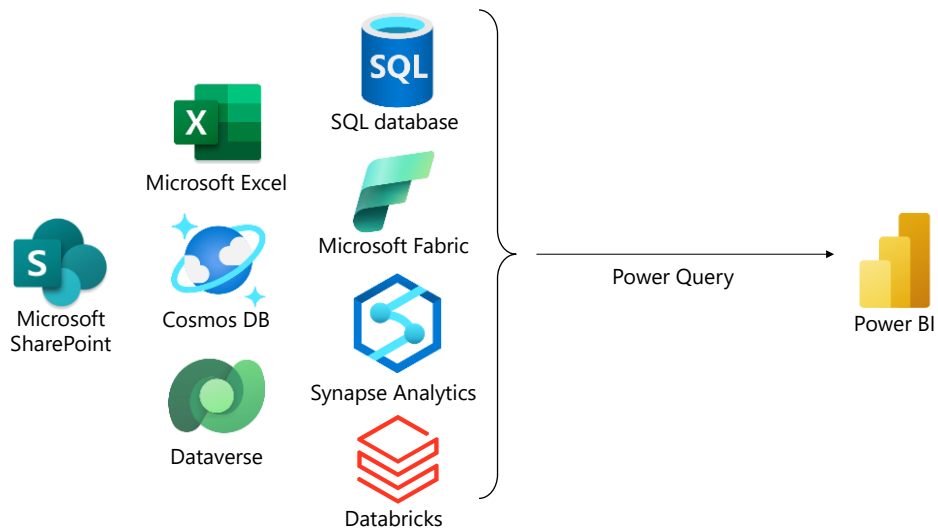
12

Get data in Power BI

© Copyright Microsoft Corporation. All rights reserved.

13

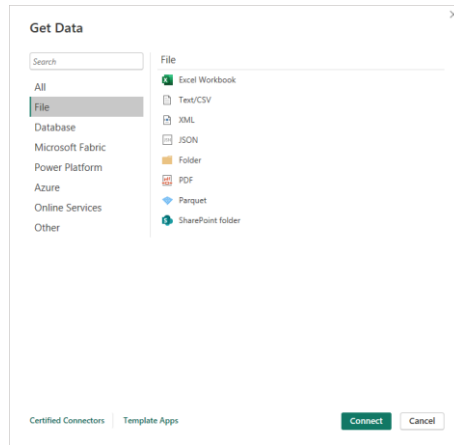
Combine all data into a single semantic model



© Copyright Microsoft Corporation. All rights reserved.

14

Get data from a wide variety of sources



© Copyright Microsoft Corporation. All rights reserved.

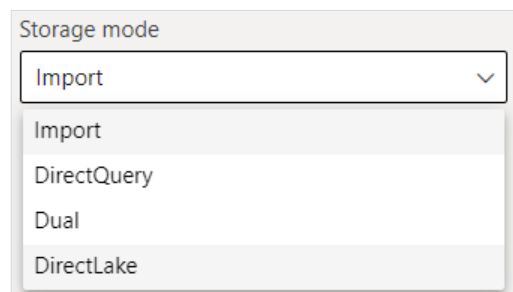
15

Select a storage mode

Storage mode affects

- Available transformations
- Report performance

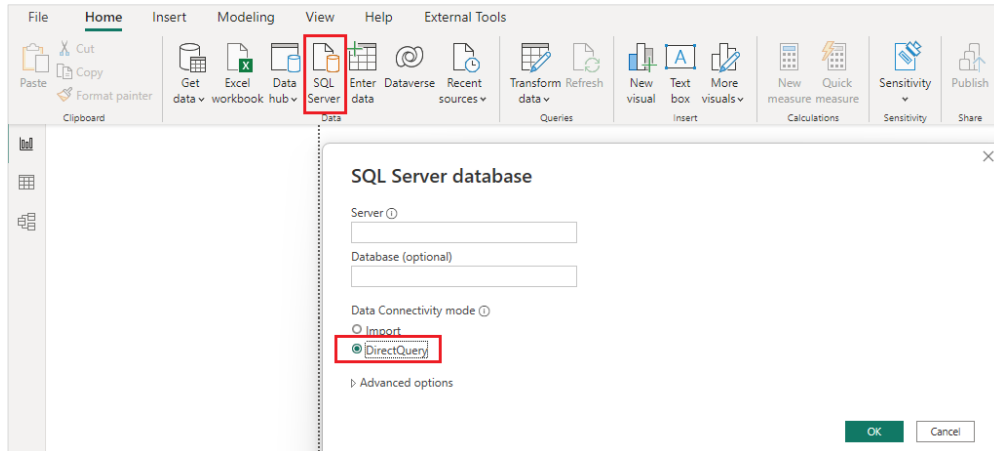
**Not all sources support all modes*



© Copyright Microsoft Corporation. All rights reserved.

16

Introduction to DirectQuery



© Copyright Microsoft Corporation. All rights reserved.

17

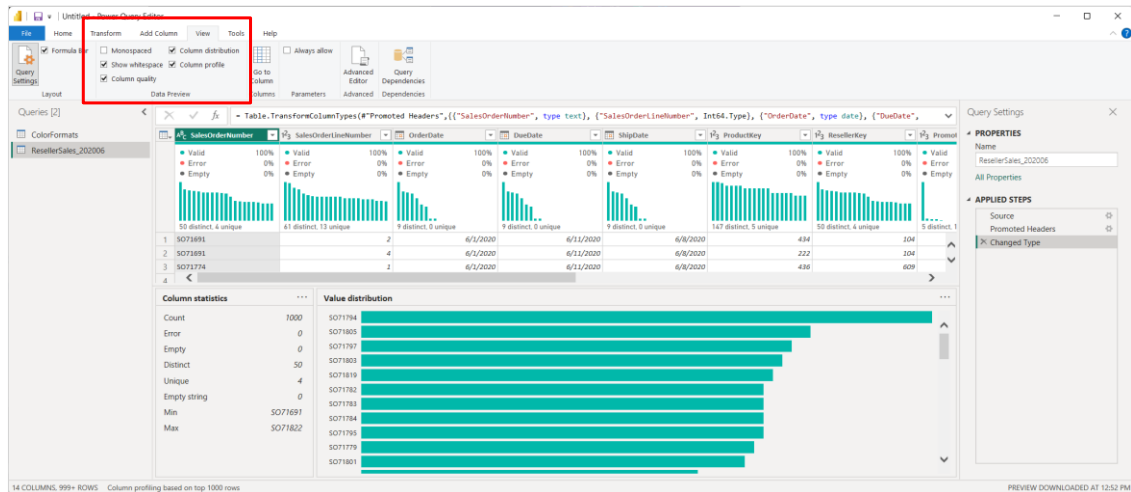
Implications of using DirectQuery

Benefits	Limitations
Frequently changing data	Dependent on data source performance
Need near real-time	Security between source and destination
Large data volumes	Limited modeling capabilities
Multi-dimensional data	Limited transformation features

© Copyright Microsoft Corporation. All rights reserved.

18

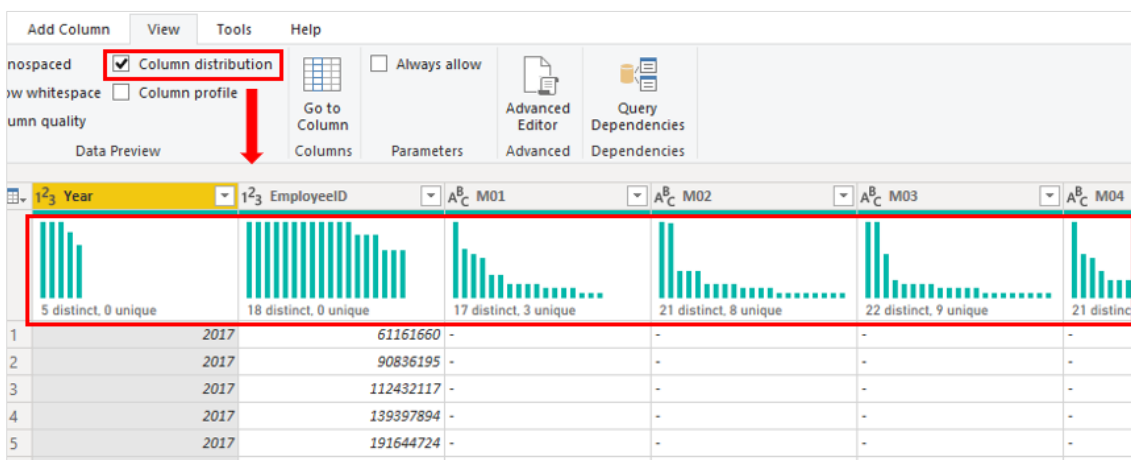
Data profiling options in Power Query Editor



© Copyright Microsoft Corporation. All rights reserved.

19

Determine cardinality of a column



© Copyright Microsoft Corporation. All rights reserved.

20

Data import errors

- Possible data load errors:
 - Query Timeout
 - Couldn't find data formatted as a table
 - Could not find file
 - Data type errors



© Copyright Microsoft Corporation. All rights reserved.

21

Further reading

Get data in Power BI

<https://learn.microsoft.com/training/modules/get-data/>



© Copyright Microsoft Corporation. All rights reserved.

22

Clean, transform, and load data

© Copyright Microsoft Corporation. All rights reserved.

23

Transform data with Power Query Editor

Table: Table.TransformColumnTypes(*Promoted Headers*, [{"SalesOrderNumber", type text}, {"SalesOrderLineNumber", Int64.Type}, {"OrderDate", ...}]

	SalesOrderNumber	SalesOrderLineNumber	OrderDate	DueDate	ShipDate	ProductKey	Row
1	SO71691	2	6/1/2020	6/11/2020	6/8/2020	434	
2	SO71691	4	6/1/2020	6/11/2020	6/8/2020	222	
3	SO71774	1	6/1/2020	6/11/2020	6/8/2020	436	
4	SO71774	2	6/1/2020	6/11/2020	6/8/2020	418	
5	SO71775	1	6/1/2020	6/11/2020	6/8/2020	573	
6	SO71775	2	6/1/2020	6/11/2020	6/8/2020	555	
7	SO71775	3	6/2/2020	6/11/2020	6/8/2020	490	
8	SO71776	1	6/2/2020	6/12/2020	6/8/2020	524	
9	SO71777	1	6/2/2020	6/12/2020	6/8/2020	408	
10	SO71777	2	6/2/2020	6/12/2020	6/8/2020	436	
11	SO71778	1	6/2/2020	6/12/2020	6/8/2020	467	
12	SO71778	2	6/2/2020	6/12/2020	6/8/2020	566	
13	SO71778	3	6/2/2020	6/12/2020	6/8/2020	554	
14	SO71778	4	6/2/2020	6/12/2020	6/8/2020	497	
15	SO71779	1	6/2/2020	6/12/2020	6/8/2020	589	
16	SO71779	2	6/2/2020	6/12/2020	6/8/2020	543	
17							

14 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

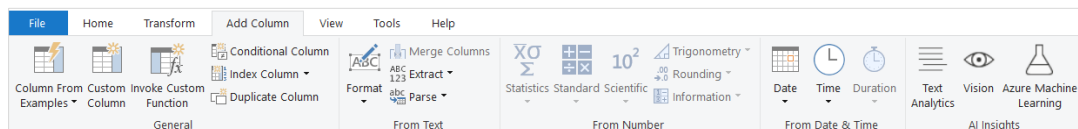
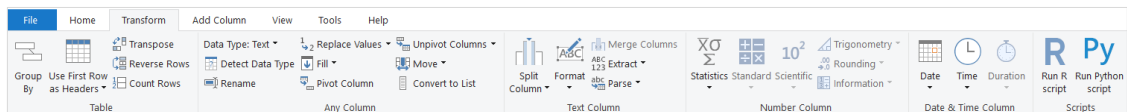
PREVIEW DOWNLOADED AT 12:52 PM

© Copyright Microsoft Corporation. All rights reserved.

24

Common transformations

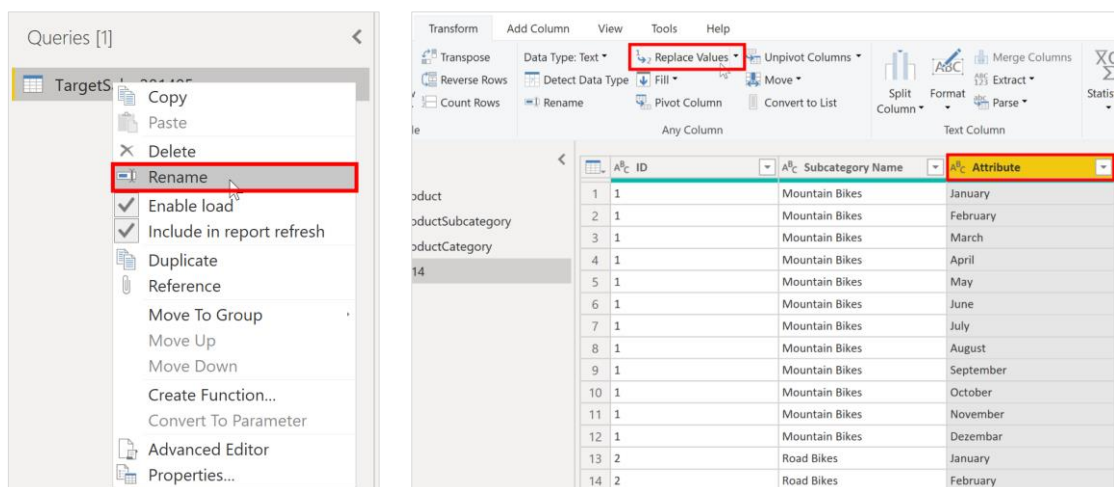
Transform columns, add new, split, extract, and more



© Copyright Microsoft Corporation. All rights reserved.

25

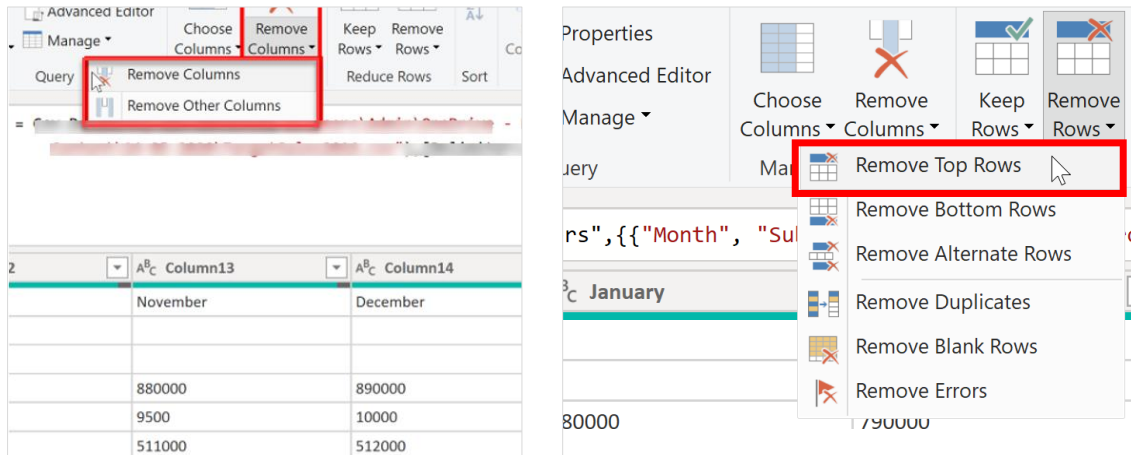
Choose user-friendly values



© Copyright Microsoft Corporation. All rights reserved.

26

Shaping table structure



© Copyright Microsoft Corporation. All rights reserved.

27

Evaluate and change column data types

ABC SalesOrderNumber	ABC OrderDate	ABC TotalCost
SO43659	7/1/2017	2024.99
SO43659	7/1/2017	6074.97
SO43659	7/1/2017	2024.99
SO43659	7/1/2017	2039.99
SO43659	7/1/2017	2039.99
SO43659	7/1/2017	4079.98
SO43659	7/1/2017	2039.99
SO43659	7/1/2017	86.52
SO43659	7/1/2017	28.84
SO43659	7/1/2017	34.2

Couldn't load the data for this visual

MdxScript(Model) (19, 40) Calculation error in measure
'Sales'[Quantity of Orders YTD]: A column specified in the call to
function 'TOTALYTD' is not of type DATE. This is not supported.

[Copy details](#)

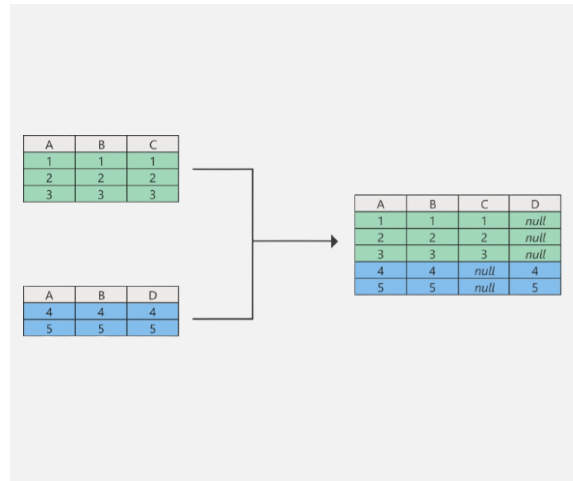
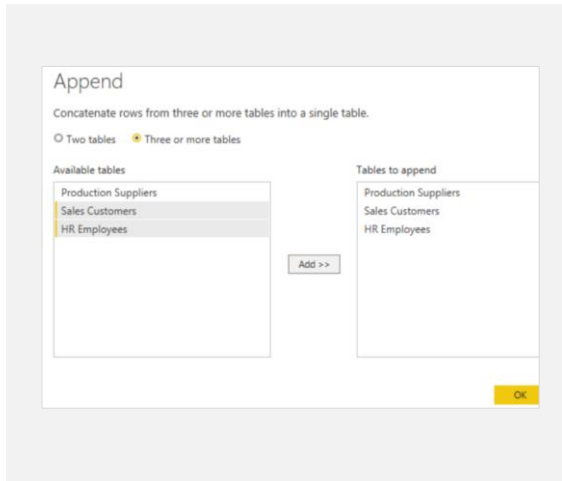
Send a Frown

Close

© Copyright Microsoft Corporation. All rights reserved.

28

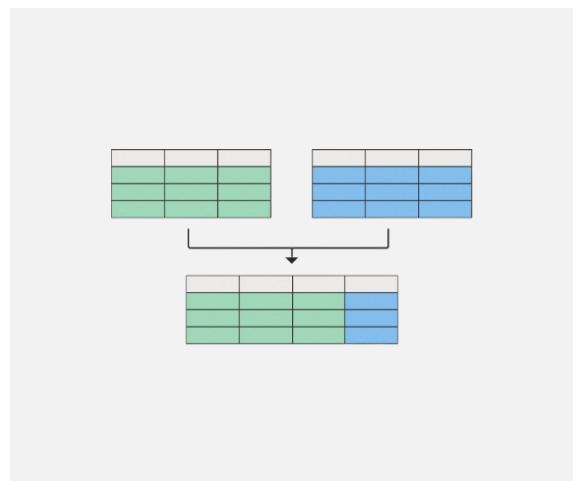
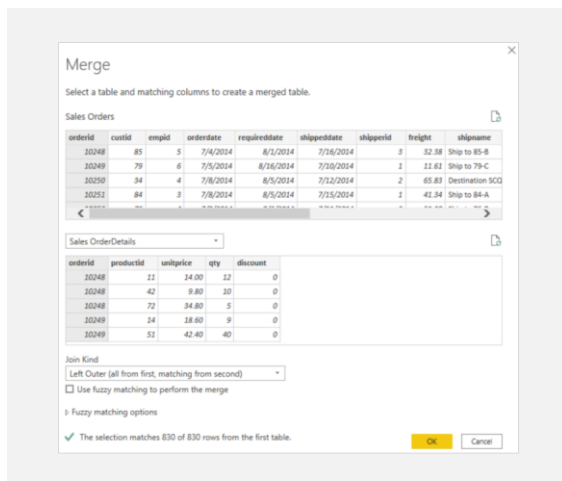
Append multiple queries into one



© Copyright Microsoft Corporation. All rights reserved.

29

Merge multiple queries into one



© Copyright Microsoft Corporation. All rights reserved.

30

Unpivot or pivot columns

Add or remove table structure to meet your aggregation needs.

	Category Name	Subcategory Name
1	Bikes	Mountain Bikes
2	Bikes	Road Bikes
3	Bikes	Touring Bikes
4	Clothing	Bib-Shorts
5	Clothing	Caps
6	Clothing	Gloves
7	Clothing	Jerseys
8	Clothing	Shorts
9	Clothing	Socks
10	Clothing	Tights
11	Clothing	Vests
12	Accessories	Bike Racks
13	Accessories	Bike Stands
14	Accessories	Bottles and Cages

	1.2 Bikes	1.2 Components	1.2 Clothing	1.2 Accessories
1	3	14	8	12

© Copyright Microsoft Corporation. All rights reserved.

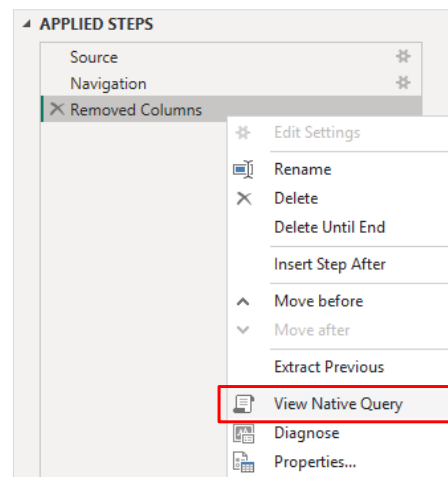
31

Query folding

Pushes data transformations to the source for better performance and efficiency.

Supported sources

- Relational databases
- OData feeds
- Active Directory



© Copyright Microsoft Corporation. All rights reserved.

32

Performance recommendations

- Only keep necessary data
- Check data types
- Reduce cardinality
- Disable query load
- Use parameters

© Copyright Microsoft Corporation. All rights reserved.

33

Further reading

Clean, transform, and load data in Power BI

<https://learn.microsoft.com/training/modules/clean-data-power-bi/>



© Copyright Microsoft Corporation. All rights reserved.

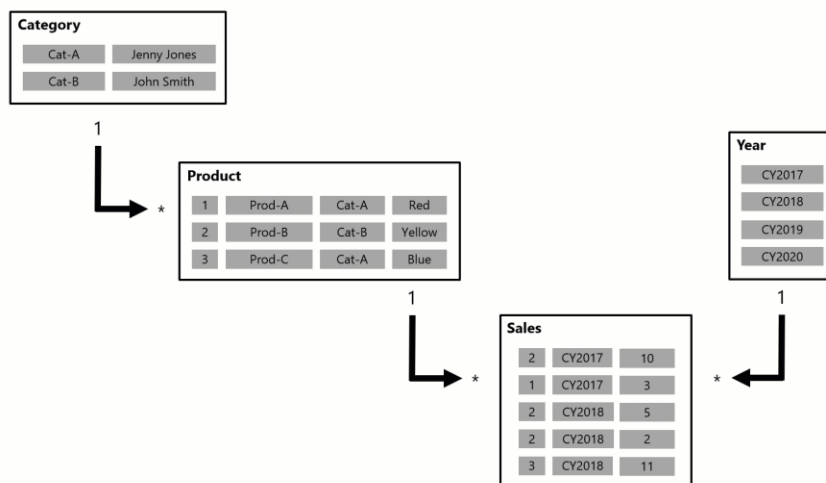
34

Configure a semantic model

© Copyright Microsoft Corporation. All rights reserved.

35

Configure relationships

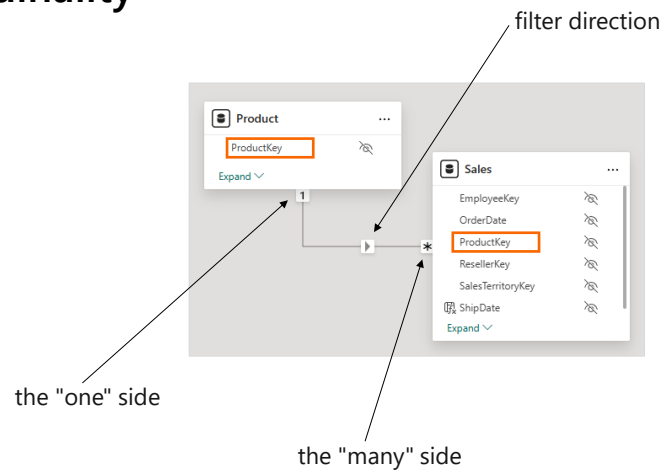


© Copyright Microsoft Corporation. All rights reserved.

36

Understand cardinality

- One-to-many (**1:***)
- Many-to-one (***:1**)
- One-to-one (**1:1**)
- Many-to-many (***:***)



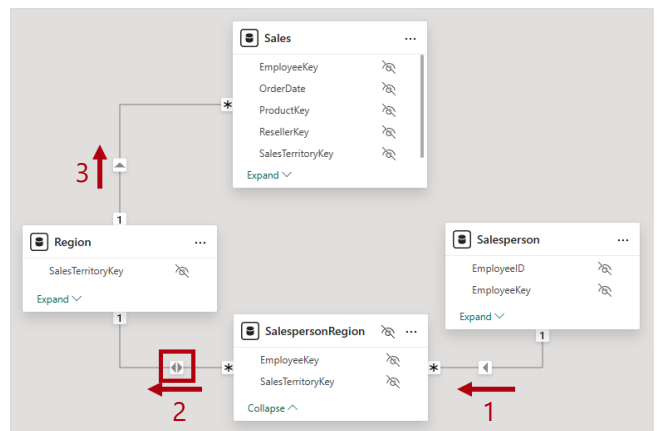
© Copyright Microsoft Corporation. All rights reserved.

37

Understand cross filter direction

The direction determines how filters propagate:

Cardinality	Cross filter direction
1:* or *:1	Single or both
1:1	Both
:	Single to either table, or both

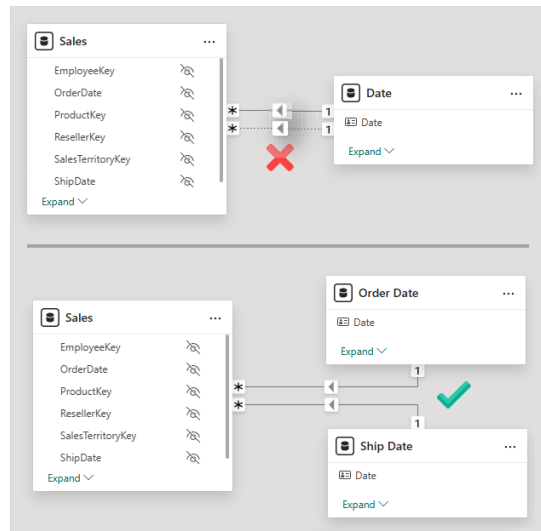


© Copyright Microsoft Corporation. All rights reserved.

38

Active vs. inactive relationships

- There can only be one active relationship between 2 tables.
- Inactive relationships can only be made active during the evaluation of a model calculation by using the USERELATIONSHIP DAX function.



© Copyright Microsoft Corporation. All rights reserved.

39

Configure table properties

- Name
- Description
- Synonyms
- Row label
- Is hidden

The Properties pane for the 'Sales' table shows the following configuration:

- Name:** Sales
- Description:** Enter a description
- Synonyms:** sale
- Row label:** Select a row label
- Key column:** Select a column with unique values
- Is hidden:** No
- Is featured table:** No

© Copyright Microsoft Corporation. All rights reserved.

- Disable auto date/time

The Time intelligence settings show the 'Auto date/time' checkbox is unchecked, indicating it is disabled. A 'Learn more' link is also present.

- Mark as data table

The 'Mark as a date table' dialog box shows the 'Mark as a date table' toggle is turned on. The 'Choose a date column' dropdown is set to 'Date'. A green checkmark indicates the operation was successful.

40

Configure columns

- General properties
- Column formatting
- Categorize data
- Set summarization (Σ)

© Copyright Microsoft Corporation. All rights reserved.

- Sort order

41

Configure Hierarchies

Relationships *between* columns

Year	Sales
FY2018	\$16,429,043
FY2018 Q1	\$3,195,733
2017 Jul	\$489,328
2017 Aug	\$1,540,072
2017 Sep	\$1,166,332
FY2018 Q2	\$4,874,023
FY2018 Q3	\$4,069,304
FY2018 Q4	\$4,289,983
FY2019	\$27,979,780
FY2020	\$33,139,748
Total	\$77,548,570

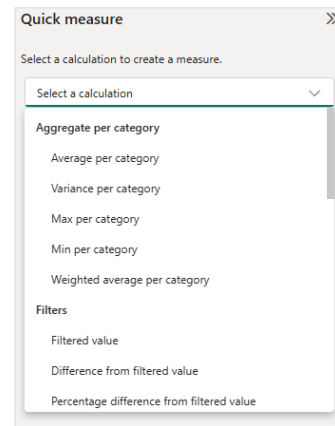
© Copyright Microsoft Corporation. All rights reserved.

42

Use Quick measures

Calculations extend the data in your semantic model

Configure properties, just like columns



© Copyright Microsoft Corporation. All rights reserved.

43

Further reading

Configure a semantic model

<https://learn.microsoft.com/training/modules/configure-semantic-model-power-bi>



© Copyright Microsoft Corporation. All rights reserved.

44

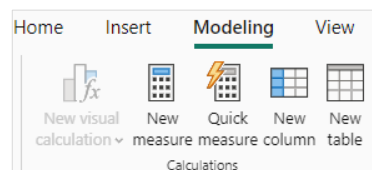
Create DAX calculations in semantic models

© Copyright Microsoft Corporation. All rights reserved.

45

What is DAX?

- **D**ata **A**nalysis **E**xpressions
- Library of functions and operators
- Build formulas and expressions
- Creates calculations: measures, columns, and tables

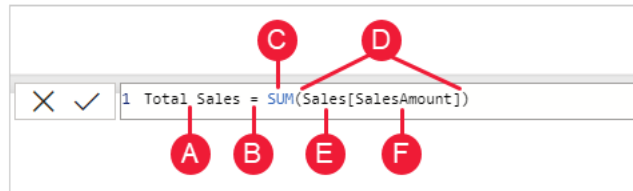


© Copyright Microsoft Corporation. All rights reserved.

46

DAX Syntax

- A. Measure name
- B. Equal sign operator
- C. DAX function
- D. Parenthesis ()
- E. Reference to table
- F. Reference to column



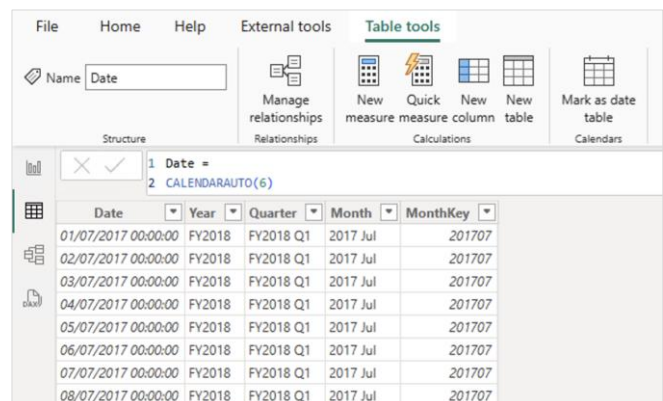
© Copyright Microsoft Corporation. All rights reserved.

47

Create calculated tables

- Defined using DAX expressions.
- Computed & stored in data model.
- Useful for aggregating data or creating custom tables.
- Table and calculator icon.

Date = CALENDARAU(6)

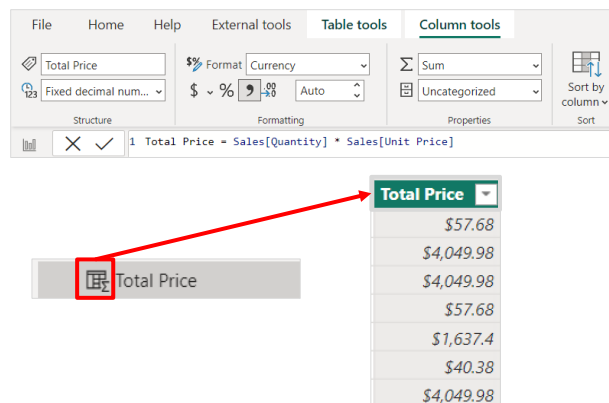


© Copyright Microsoft Corporation. All rights reserved.

48

Create calculated columns

- Calculated for each row in the table.
- Computed & stored in data model.
- Useful “helper/connector columns.”
- Recalculated during data refresh.
- Table and Sigma icon.



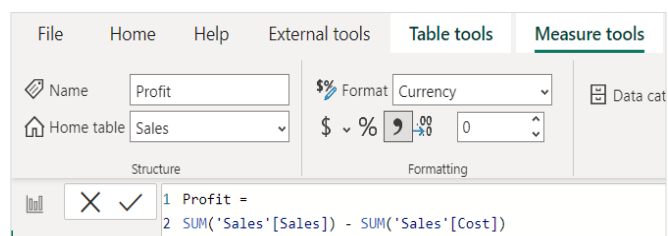
Total Price = Sales[Quantity] * Sales[Unit Price]

© Copyright Microsoft Corporation. All rights reserved.

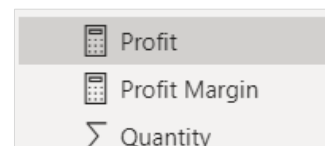
49

Create measures

- Defined with DAX definitions.
- Computed on the fly.
- Not stored in data model.
- Responsive to interactions.
- Indicated by calculator icon.



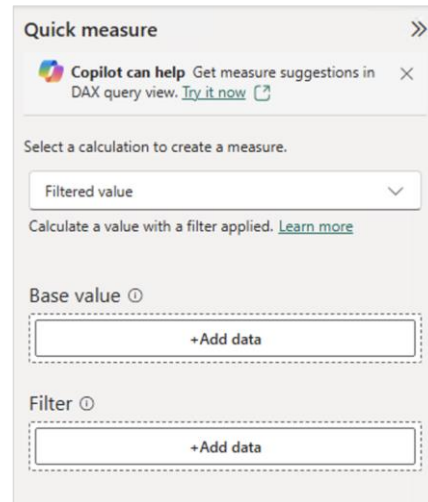
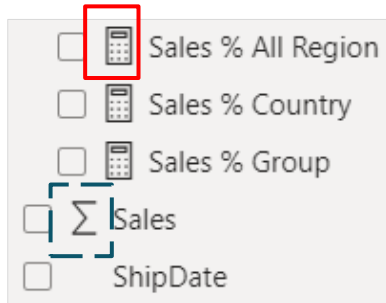
Profit = SUM('Sales'[Sales]) - SUM('Sales'[Cost])



© Copyright Microsoft Corporation. All rights reserved.

50

Implicit vs. Explicit vs. Quick measures



© Copyright Microsoft Corporation. All rights reserved.

51

Common DAX functions

Function	Descriptions
SUM()	Adds up values
MIN(), MAX()	Smallest/largest value
FILTER()	Returns a filtered table
CALCULATE()	Core DAX function to change context
ALL(), REMOVEFILTERS()	Remove filters
FORMAT()	Format numbers/dates

```
TotalSales = SUM(Sales[Amount])
TotalCost = SUM(Sales[Cost])
Profit = [TotalSales] - [TotalCost]
ProfitMargin = DIVIDE([Profit], [TotalSales])
```

} Compound measure
(promotes reuse)

© Copyright Microsoft Corporation. All rights reserved.

52

Iterator functions

SUMX, COUNTX, AVERAGEX, MINX, MAXX, ...

Useful for:

- Complex summarizations
- Higher grain summarization
- Ranking

iterate over each sales order line (higher grain summarization)

```
Revenue Avg Order =
AVERAGEX(
    VALUES('Sales Order'[Sales Order]),
    [Revenue]
)
```

Month	Revenue	Discount	Revenue Avg Order Line	Revenue Avg Order
2017 Jul	\$1,423,357.32	\$326,219.06	\$2,220.53	\$4,352.77
2017 Aug	\$2,057,902.45	\$1,031,506.86	\$2,179.98	\$8,794.45
2017 Sep	\$2,523,947.55	\$1,342,355.67	\$2,014.32	\$9,670.30
2017 Oct	\$561,681.48	\$0.00	\$3,228.05	\$3,228.05
2017 Nov	\$4,764,920.16	\$2,692,205.61	\$2,227.64	\$12,441.04
2017 Dec	\$596,746.56	\$0.00	\$3,174.18	\$3,174.18
2018 Jan	\$1,327,674.63	\$475,813.07	\$2,329.25	\$5,698.17
2018 Feb	\$3,936,463.31	\$2,237,412.77	\$2,321.03	\$12,301.45
2018 Mar	\$700,873.18	\$0.00	\$3,200.33	\$3,200.33
2018 Apr	\$1,519,275.24	\$588,995.26	\$2,182.87	\$6,356.80
2018 May	\$2,960,378.09	\$1,514,891.49	\$2,219.17	\$9,642.93
2018 Jun	\$1,487,671.19	\$1,659,893.12	\$1,398.19	\$4,752.94

© Copyright Microsoft Corporation. All rights reserved.

53

Calculated columns vs. measures

	Calculated columns	Measures
Purpose	Adds a new column to a table	Defines how to summarize model data
Evaluation	Evaluated using row context at data refresh time	Evaluated using filter context at query time
Storage	Stores a value for each row in the table (Import mode)	Never stores values in the model
Use in visuals	Can filter, group, or summarize data (as implicit measures)	Designed specifically to summarize data
Performance impact	Can increase memory usage and model size	More efficient; better performance in large models
Ideal for	New fields for slicing or relationships	Dynamic calculations based on filters

© Copyright Microsoft Corporation. All rights reserved.

54

Further reading

Write DAX formulas for Power BI

<https://learn.microsoft.com/training/modules/dax-power-bi-write-formulas>

Create DAX calculations in semantic models

<https://learn.microsoft.com/training/modules/dax-power-bi-create-calculations/>



© Copyright Microsoft Corporation. All rights reserved.

55

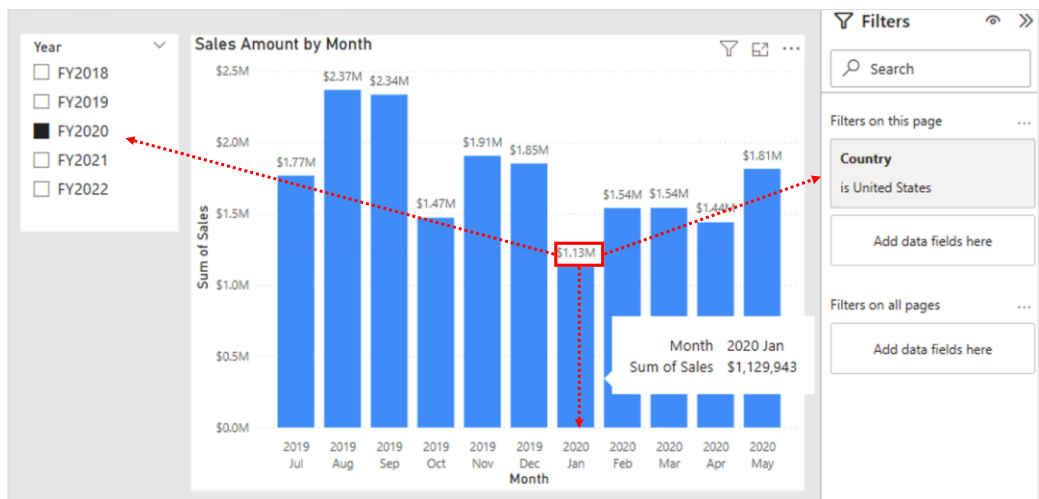
Modify DAX filter context

© Copyright Microsoft Corporation. All rights reserved.

56

Understanding filter context

Measures are “dynamic,” depending on filters.



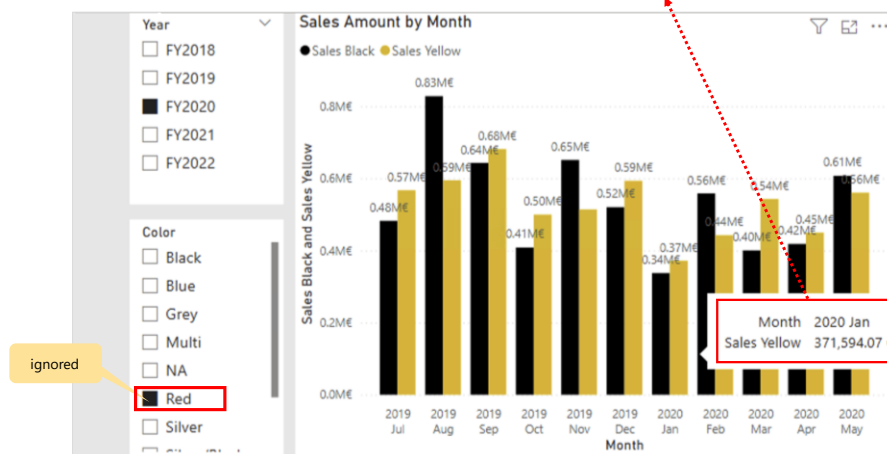
© Copyright Microsoft Corporation. All rights reserved.

57

Modify filter context using CALCULATE()

Sales Black = `CALCULATE(SUM(Sales[Sales]), 'Product'[Color]="Black")`

Sales Yellow = `CALCULATE(SUM(Sales[Sales]), 'Product'[Color]="Yellow")`



© Copyright Microsoft Corporation. All rights reserved.

58

Further reading

Modify DAX filter context in semantic models

<https://learn.microsoft.com/training/modules/dax-power-bi-modify-filter>

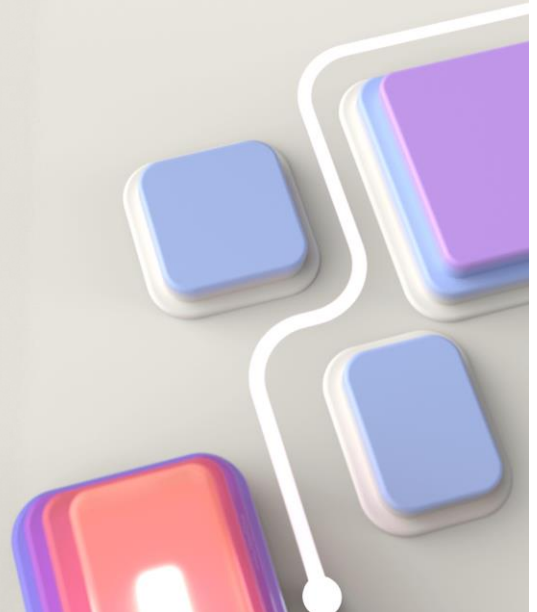


© Copyright Microsoft Corporation. All rights reserved.

61

Use DAX time intelligence functions

© Copyright Microsoft Corporation. All rights reserved.



62

Create a common date table

Mark as date table

×

Select a column to be used for the date. The column must be of the data type 'date' and must contain only unique values. [Learn more](#)

Date column

Date

▼

✓ Validated successfully

When you mark this as a date table, the built-in date tables that were associated with this table are removed. Visuals or DAX expressions referring to them may break. [Learn how to fix visuals and DAX expressions](#)

OK

Cancel

Mark as date table

- Contain unique values.
- Span full years.
- Not contain BLANKs.
- Not have any missing dates.

© Copyright Microsoft Corporation. All rights reserved.

63

Use DAX time intelligence functions

Summarizations over time

- TOTALYTD, TOTALQTD, TOTALMTD
- DATESBETWEEN
- ...

Comparisons over time

- DATEADD
- SAMEPERIODLASTYEAR
- PREVIOUSQUARTER, PREVIOUSYEAR
- ...

Total Sales Previous Month =
`CALCULATE(SUM(Sales[Total Price]),
PREVIOUSMONTH('Date'[Date]))`

Year	Month	Total Sales	Total Sales Previous Month
2015	March	\$39,979.9	\$41,207.2
2015	April	\$55,699.39	\$39,979.9
2015	May	\$56,823.7	\$55,699.39
2015	June	\$39,088	\$56,823.7
2015	July	\$55,464.93	\$39,088
2015	August	\$49,981.69	\$55,464.93
2015	September	\$59,733.02	\$49,981.69
2015	October	\$70,328.5	\$59,733.02
2015	November	\$45,913.36	\$70,328.5

© Copyright Microsoft Corporation. All rights reserved.

64

Point in time calculations

Stock on Hand =
`CALCULATE(SUM(Inventory[UnitsBalance]),
 LASTDATE('Date'[Date]))`

LASTDATE() Returns the last date in the current context

111 units on stock on Nov 30, 2019

FY2020 Mountain-200 Bike Stock

Product	2019 Jul	2019 Aug	2019 Sep	2019 Oct	2019 Nov	2019 Dec	2020 Jan	2020 Feb	2020 Mar	2020 Apr	2020 May
Mountain-200 Black, 38	151	171	99	172	30	168	168	168	168	168	168
Mountain-200 Black, 42	165	186	116	176	76	167	167	167	167	167	167
Mountain-200 Black, 46	182	184	131	172	111	190	190	190	190	190	190
Mountain-200 Silver, 38	171	173	129	190	85	190	190	190	190	190	190
Mountain-200 Silver, 42	177	169	109	170	120	162	162	162	162	162	162
Mountain-200 Silver, 46	181	158	126	178	88	163	163	163	163	163	163
Total	1,027	1,041	710	1,058	510	1,040	1,040	1,040	1,040	1,040	1,040

© Copyright Microsoft Corporation. All rights reserved.

65

Further reading

Use DAX time intelligence functions in semantic models

<https://learn.microsoft.com/training/modules/dax-power-bi-time-intelligence/>



© Copyright Microsoft Corporation. All rights reserved.

66

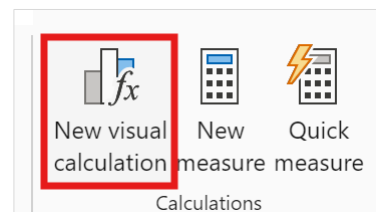
Create visual calculations

© Copyright Microsoft Corporation. All rights reserved.

67

What are visual calculations?

- DAX calculations
- Stored on the visual
- Visual representation of calculations



Compared to measures that are stored in the semantic model and must be added to a visual.

© Copyright Microsoft Corporation. All rights reserved.

68

Explore the visual calculation interface



© Copyright Microsoft Corporation. All rights reserved.

69

Create a visual calculation

- Start with a visual
- Add new visual calculation
- Add a calculation in formula bar, such as:

✕ ✓ fx 1 Profit = [Sales Amount] - [Total Product Cost]			
Fiscal Year	Sales Amount	Total Product Cost	Profit
FY2018	\$23,860,891.17	\$20,824,957.60	3,035,933.57
FY2019	\$34,070,108.50	\$30,362,108.34	3,708,000.16
FY2020	\$51,878,274.54	\$46,070,842.02	5,807,432.52
Total	\$109,809,274.20	\$97,257,907.95	12,551,366.25

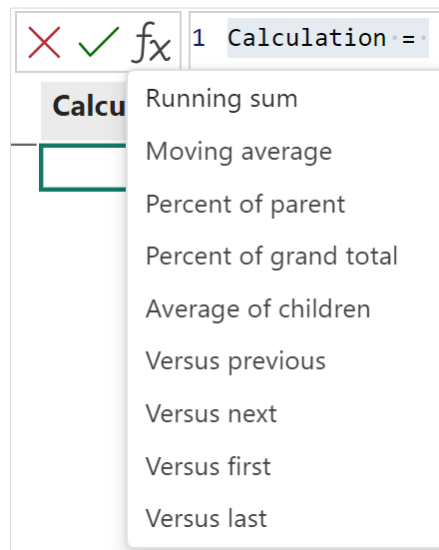
Profit = [Sales Amount] - [Total Product Cost]

© Copyright Microsoft Corporation. All rights reserved.

70

Use templates

Templates reduce code complexity by using a single function to perform complex calculations.



© Copyright Microsoft Corporation. All rights reserved.

71

Hide visual calculations

- Hide fields in the visual matrix when:
- The visual calculation depends on the field.
- The field doesn't make sense for your visual.



© Copyright Microsoft Corporation. All rights reserved.

72

Use the Axis parameter

- Determines evaluation direction

Parameters	Icon
ROWS	⇅
COLUMNS	⇆
ROWS COLUMNS	↻
COLUMNS ROWS	↺

© Copyright Microsoft Corporation. All rights reserved.

73

Use the Reset parameter

- Doesn't reset by default
- HIGHESTPARENT
- LOWESTPARENT



© Copyright Microsoft Corporation. All rights reserved.

74

Further reading

Create visual calculations in Power BI Desktop

<https://learn.microsoft.com/training/modules/power-bi-visual-calculations/>



© Copyright Microsoft Corporation. All rights reserved.

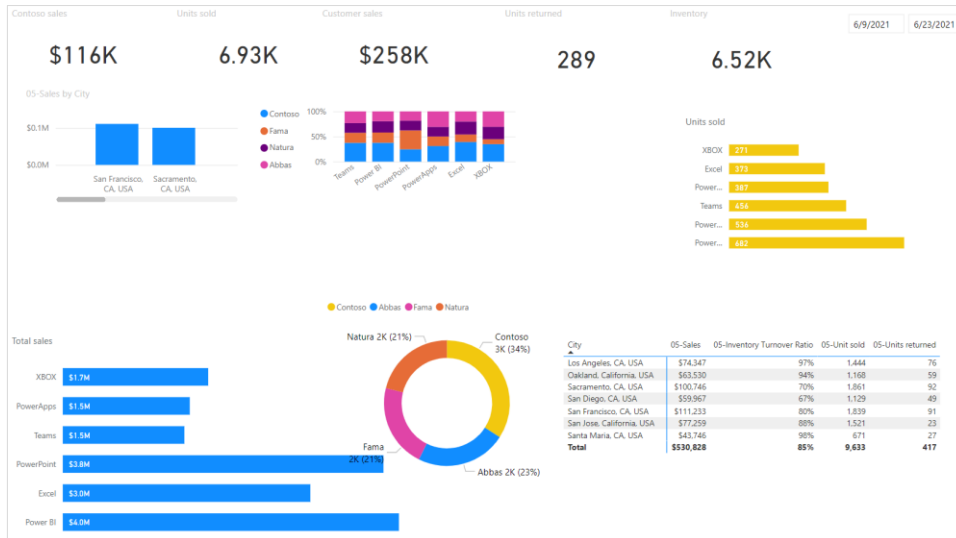
75

Design Power BI reports

© Copyright Microsoft Corporation. All rights reserved.

76

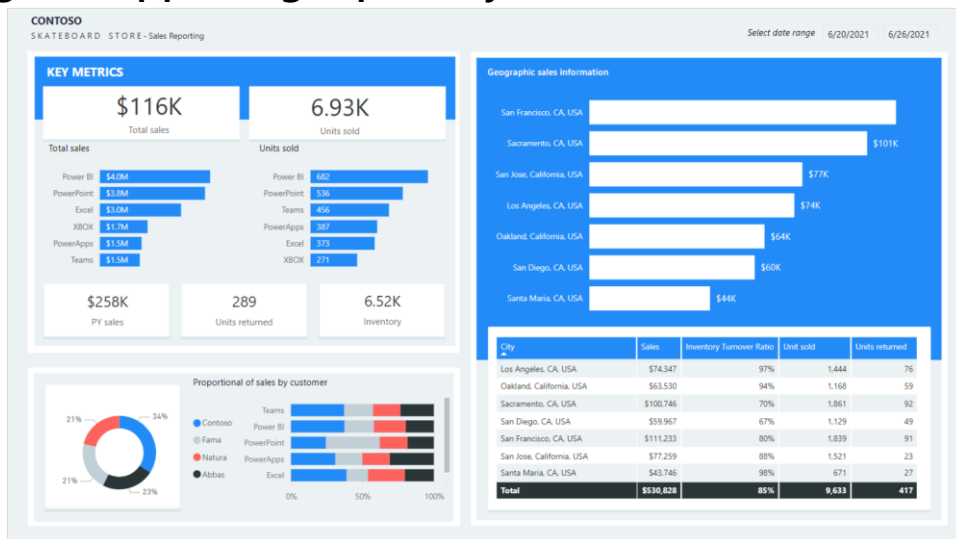
What do you think about this report?



© Copyright Microsoft Corporation. All rights reserved.

77

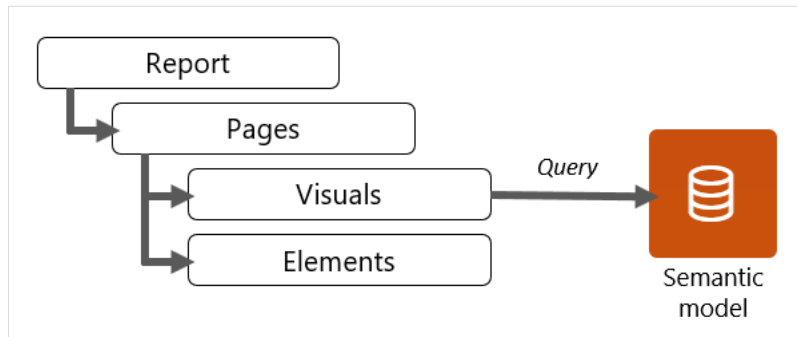
Design an appealing report layout



© Copyright Microsoft Corporation. All rights reserved.

78

Report structure

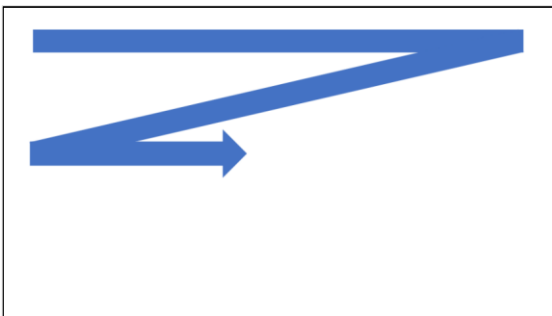


© Copyright Microsoft Corporation. All rights reserved.

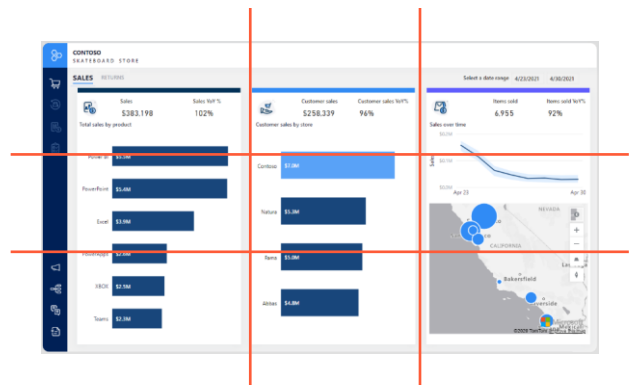
79

Design the report layout

Placement



Rule of thirds



© Copyright Microsoft Corporation. All rights reserved.

80

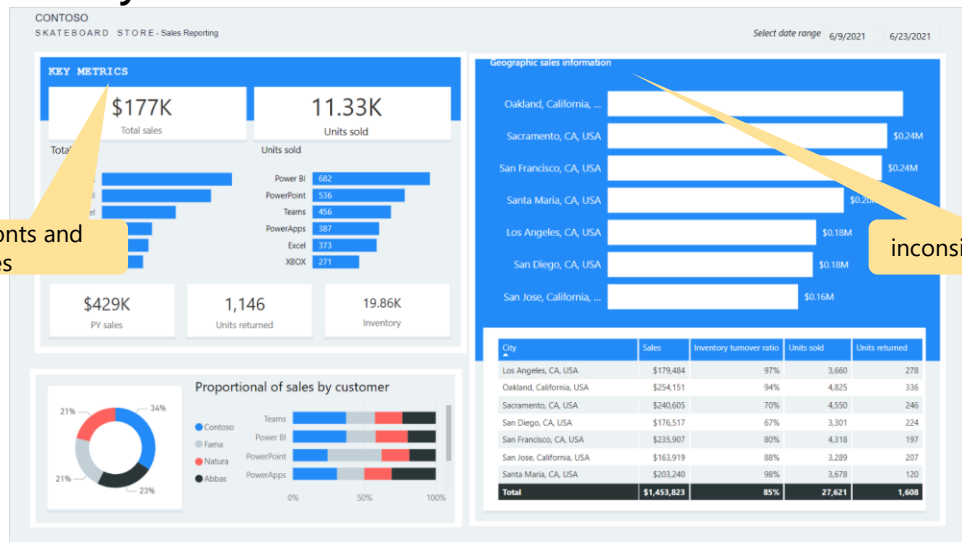
Margins and object spacing



© Copyright Microsoft Corporation. All rights reserved.

81

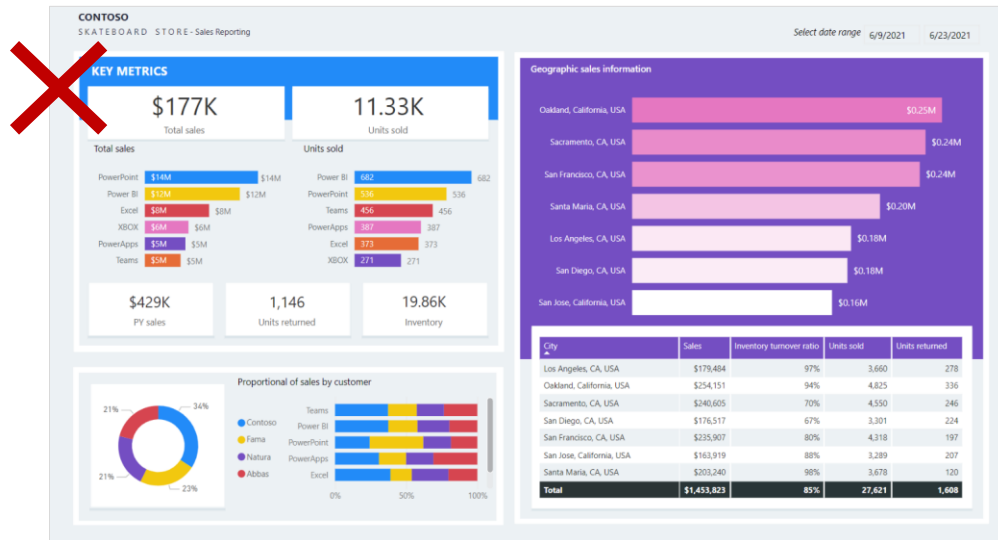
Consistency



© Copyright Microsoft Corporation. All rights reserved.

82

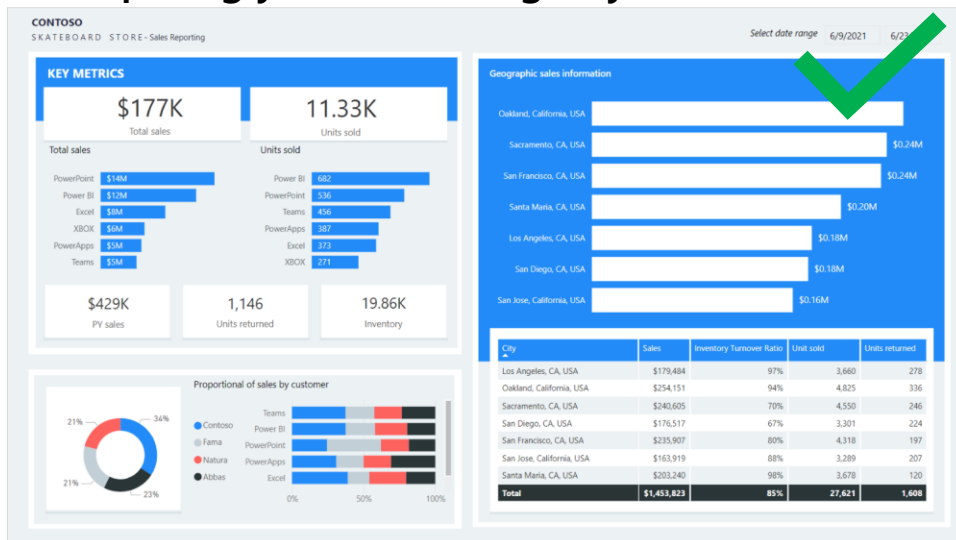
Use color sparingly and meaningfully



© Copyright Microsoft Corporation. All rights reserved.

83

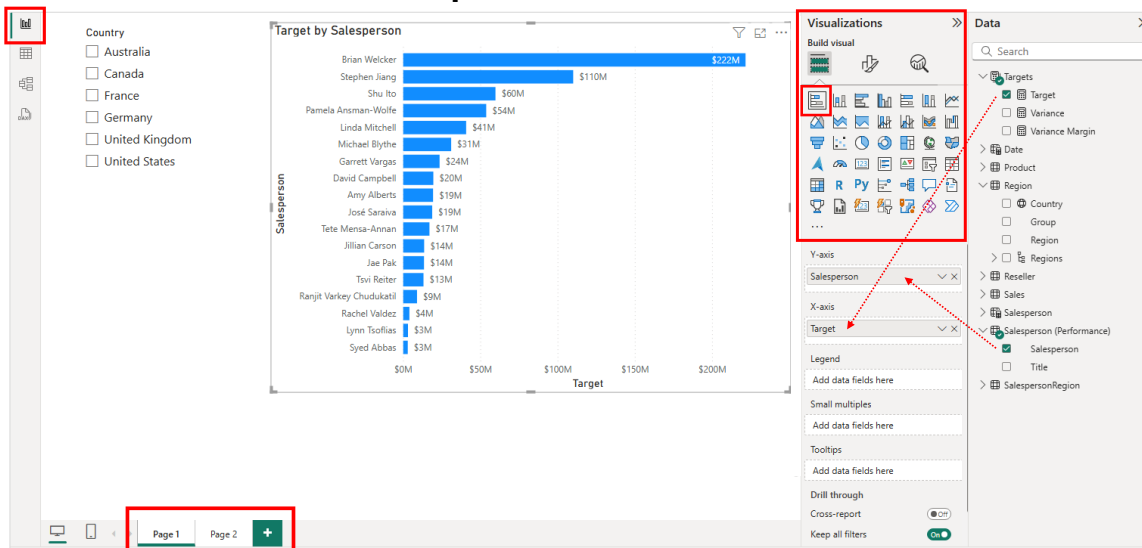
Use color sparingly and meaningfully



© Copyright Microsoft Corporation. All rights reserved.

84

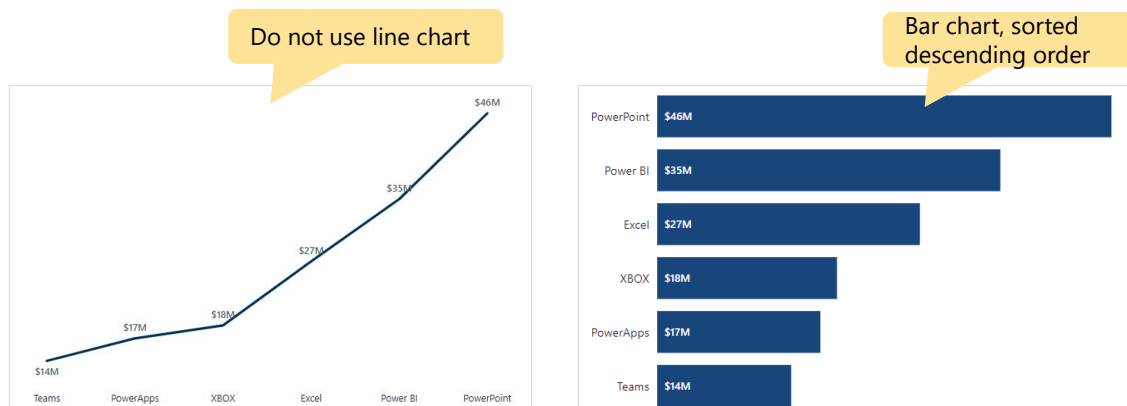
Add visualizations to reports



© Copyright Microsoft Corporation. All rights reserved.

85

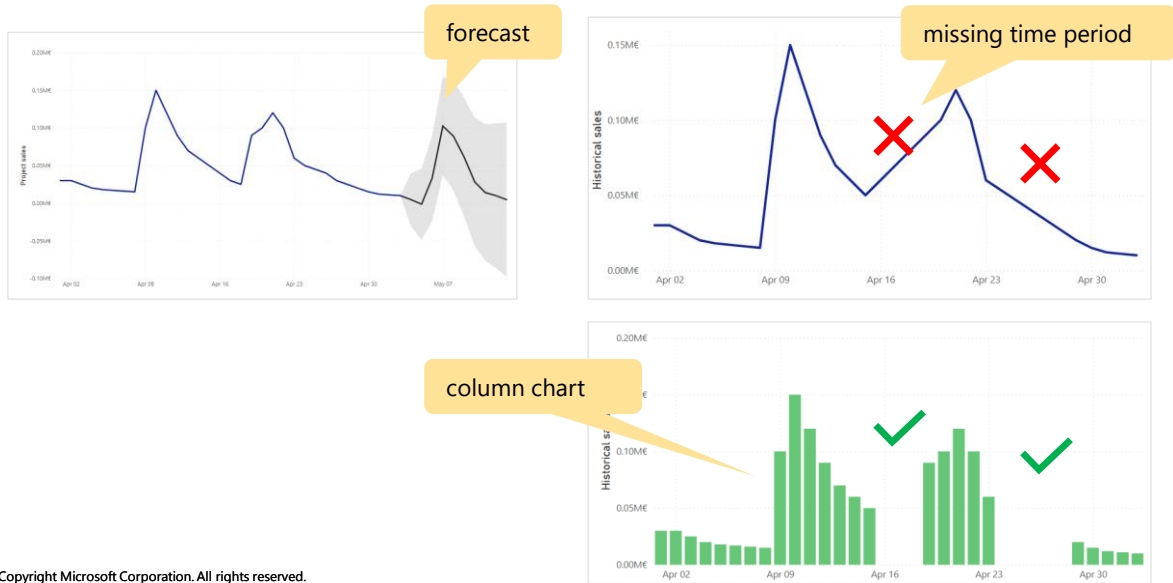
Categorical visuals



© Copyright Microsoft Corporation. All rights reserved.

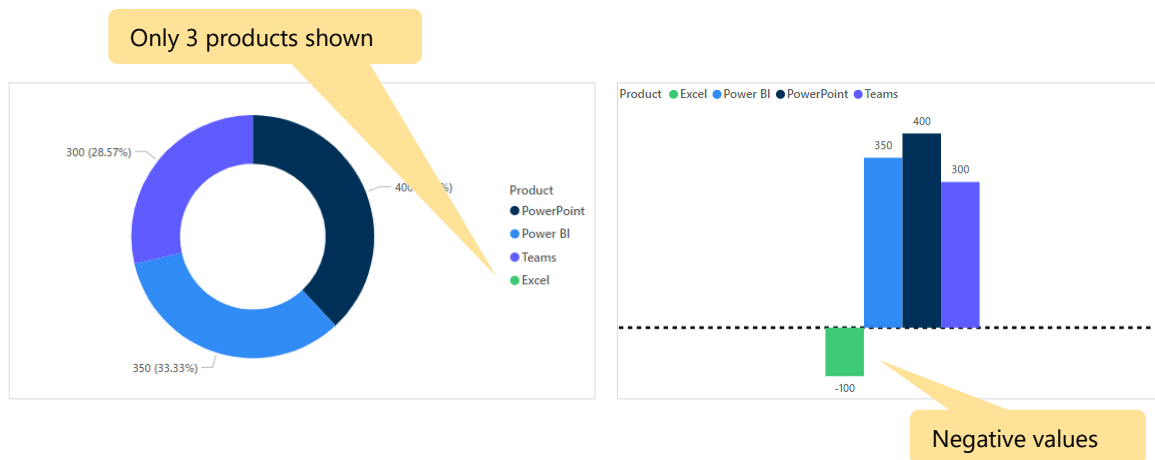
86

Time series visuals



87

Proportional visuals



88

Grid visuals

table

Product	Sales	Units sold
Power BI	\$244,885	2,881
PowerPoint	\$172,320	2,154
Excel	\$102,250	2,045
Teams	\$88,144	3,148
PowerApps	\$80,496	2,236
XBOX	\$74,400	1,860
Total	\$762,495	14,324

matrix

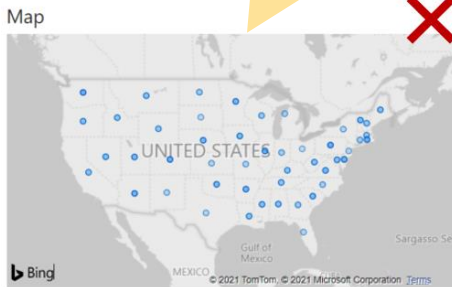
Inventory on hand breakdown			
City	On hand	Inventory turnover ratio	Days sales of inventory
Los Angeles, CA, USA	32	100% ★	✓
Power BI	2 ○	100% ★	✓
Excel	0 ○	100% ★	⚠
PowerPoint	0 ○	100% ★	⚠
Teams	30 ○	99% ★	✓
PowerApps	0 ○	100% ★	⚠
Total	1,706	97% ★	✓

© Copyright Microsoft Corporation. All rights reserved.

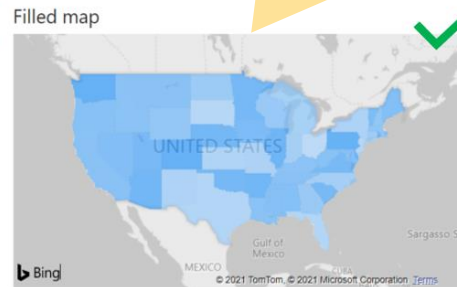
89

Geospatial visuals

map, with bubbles



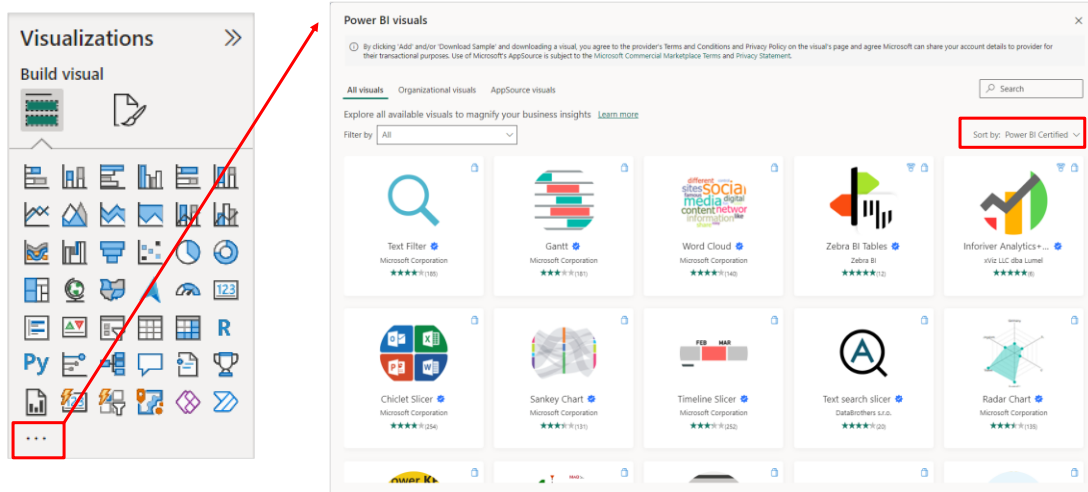
filled map, color graduations



© Copyright Microsoft Corporation. All rights reserved.

90

Import custom visuals from AppSource

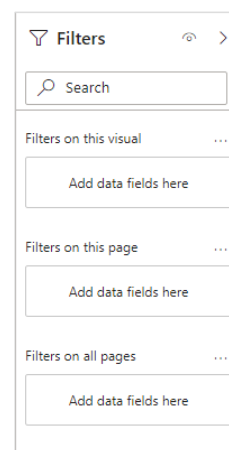
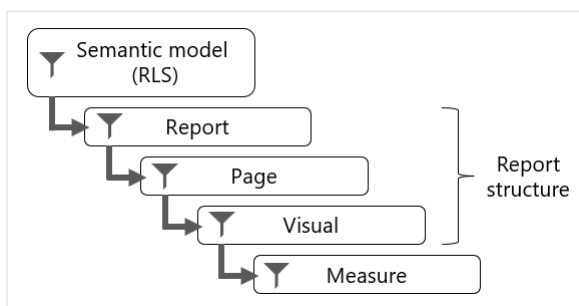


© Copyright Microsoft Corporation. All rights reserved.

91

Report filtering

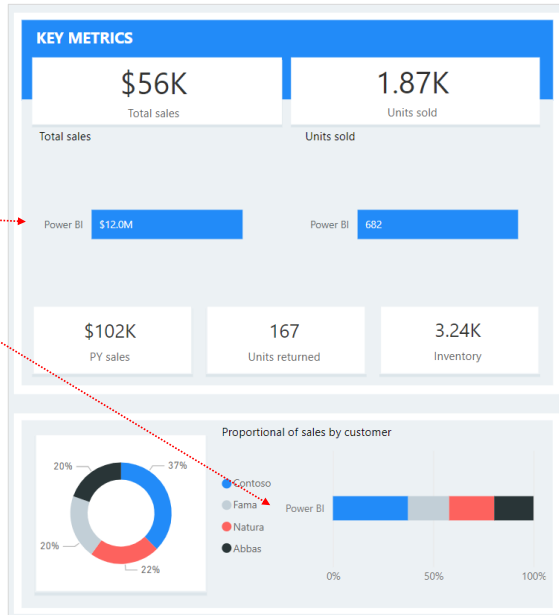
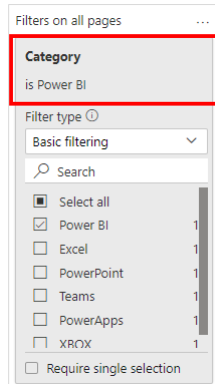
Filtering can occur at five different levels of a report.



© Copyright Microsoft Corporation. All rights reserved.

92

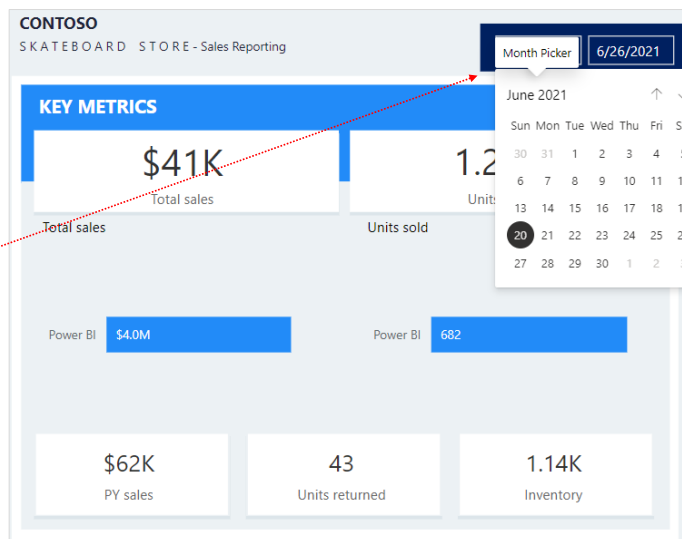
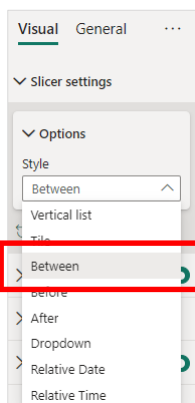
Apply filters to the report



© Copyright Microsoft Corporation. All rights reserved.

93

Apply filters with slicers



© Copyright Microsoft Corporation. All rights reserved.

94

Further reading

Design Power BI reports

<https://learn.microsoft.com/training/modules/power-bi-effective-reports/>

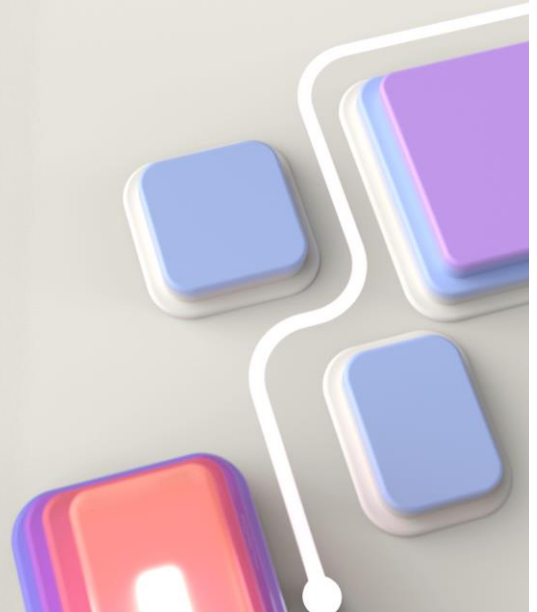


© Copyright Microsoft Corporation. All rights reserved.

95

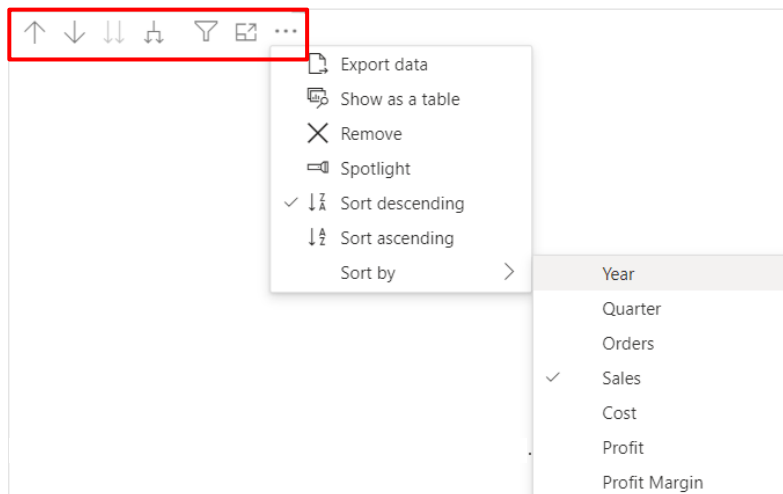
Enhance reports for user experience

© Copyright Microsoft Corporation. All rights reserved.



96

Sort visuals in the most effective order



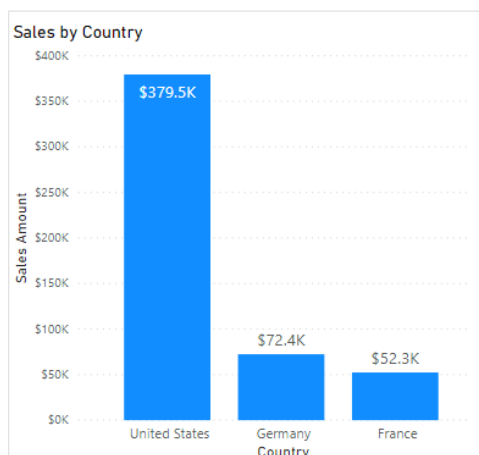
© Copyright Microsoft Corporation. All rights reserved.

97

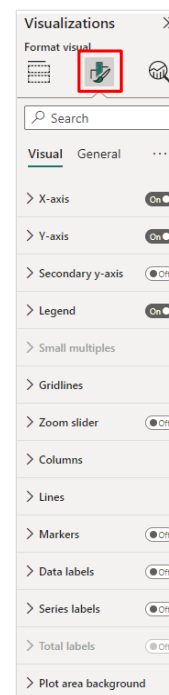
Format and configure visualizations

Common options:

- Title
- Background
- Border
- Font type/size/color
- Data labels
- Tooltips



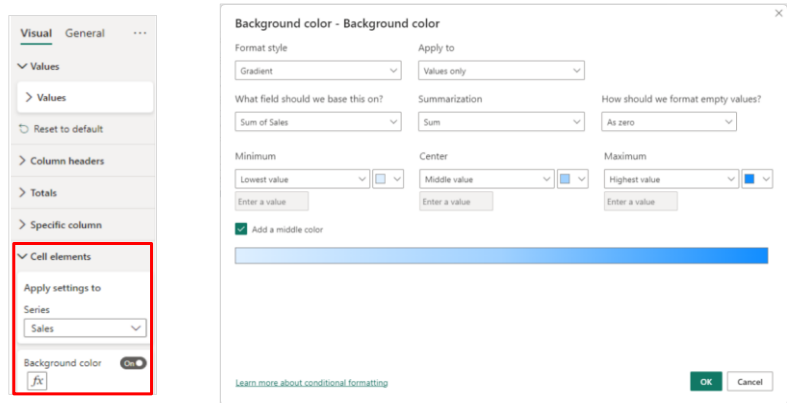
© Copyright Microsoft Corporation. All rights reserved.



98

Configure conditional formatting

Enhance certain visuals with dynamic color schemes.

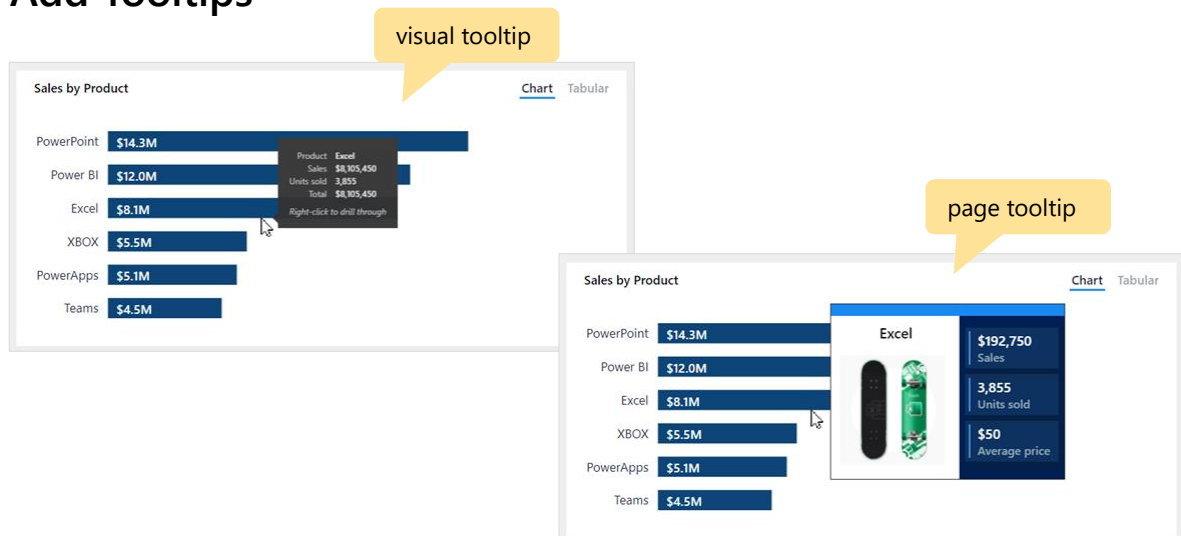


Region	Sales
Australia	\$1,391,025
Canada	\$13,875,633
Central	\$7,633,387
France	\$4,527,840
Germany	\$1,877,743
Northeast	\$6,715,354
Northwest	\$12,004,822
Southeast	\$7,638,607
Southwest	\$18,001,116
United Kingdom	\$3,883,043
Total	\$77,548,570

© Copyright Microsoft Corporation. All rights reserved.

99

Add Tooltips



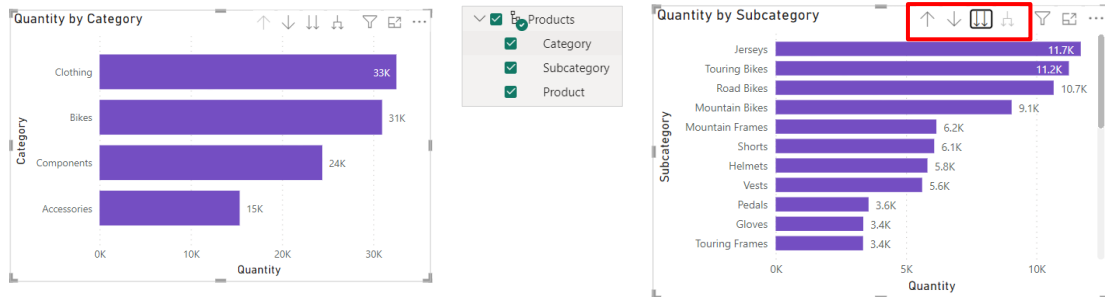
visual tooltip

page tooltip

© Copyright Microsoft Corporation. All rights reserved.

100

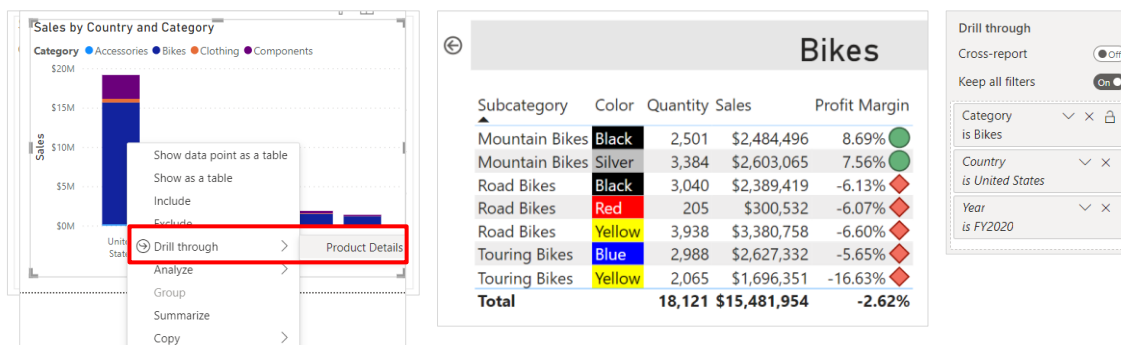
Drill down through a hierarchy in a visual



© Copyright Microsoft Corporation. All rights reserved.

101

Drill through from one visual to another

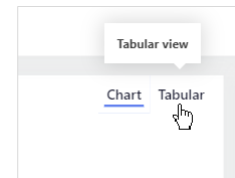
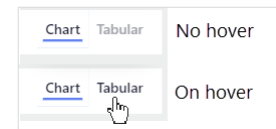
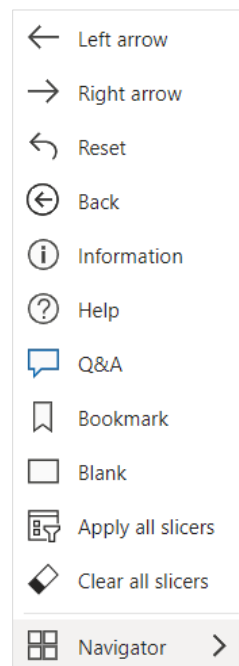


© Copyright Microsoft Corporation. All rights reserved.

102

Add shapes and buttons

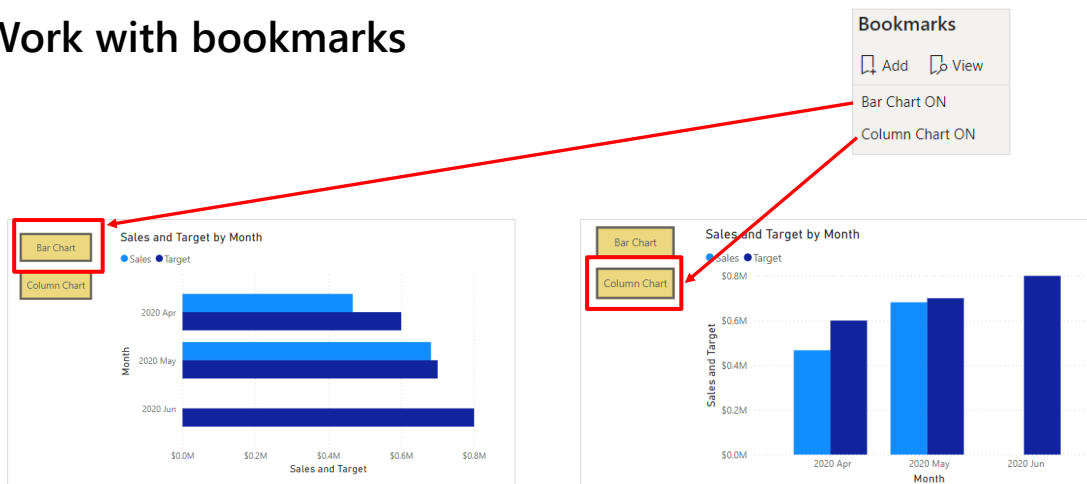
- Enhance reports with:
 - Text boxes
 - Images
 - Shapes
- Improve navigation with Buttons.



© Copyright Microsoft Corporation. All rights reserved.

103

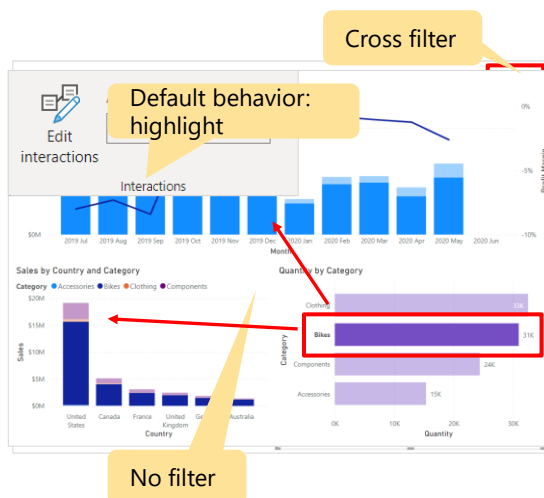
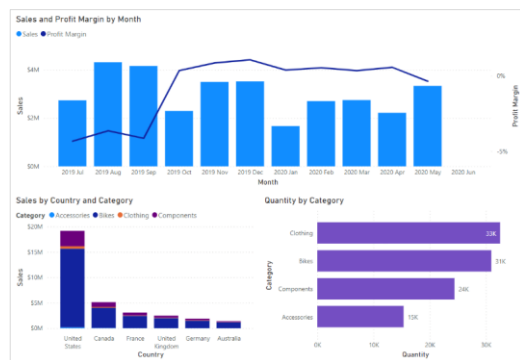
Work with bookmarks



© Copyright Microsoft Corporation. All rights reserved.

104

Cross-highlight between visuals



© Copyright Microsoft Corporation. All rights reserved.

105

Sync slicers

Sync slicers >> X

Add and sync with all pages, or select specific pages:

Page name

Page name	Sync	Filter
Overview	☒	☒
Profit	☒	☒
My Performance	☐	☐
Product Details	☐	☐

Advanced options

Enter a group name to sync selection to any other visuals with that group name

Region

☒ Sync field changes to other slicers

☒ Sync filter changes to other slicers

Year

FY2020

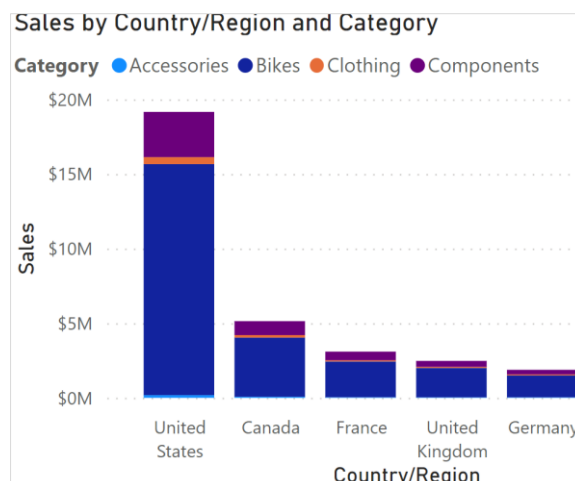
Group, Country, Region

☐ Select all

☒ Europe

☒ North America

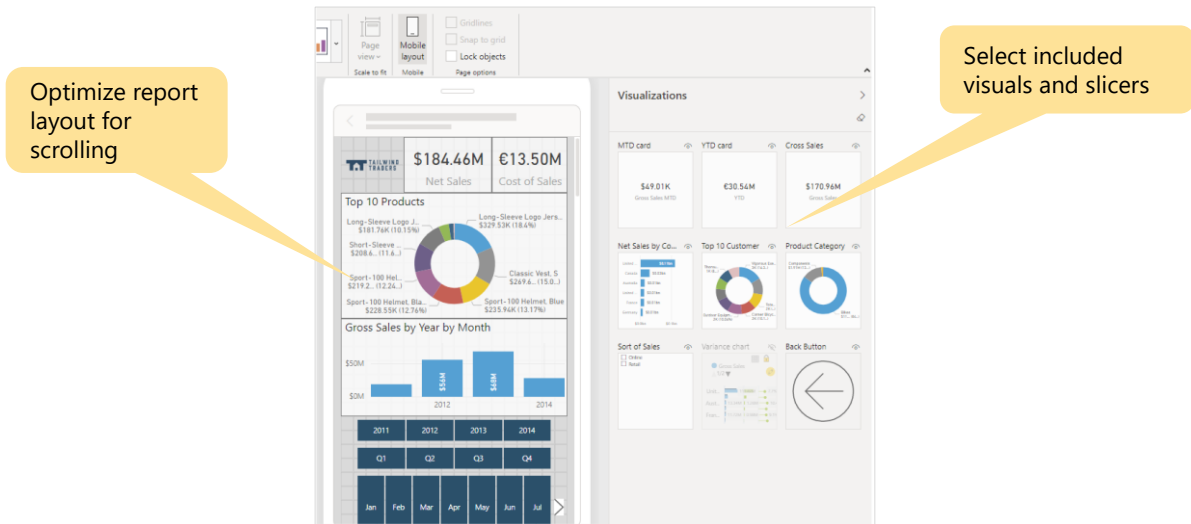
☐ Pacific



© Copyright Microsoft Corporation. All rights reserved.

106

Design for mobile use



© Copyright Microsoft Corporation. All rights reserved.

107

Further reading

Enhance Power BI report designs for the user experience

<https://learn.microsoft.com/training/modules/power-bi-effective-user-experience/>



© Copyright Microsoft Corporation. All rights reserved.

108

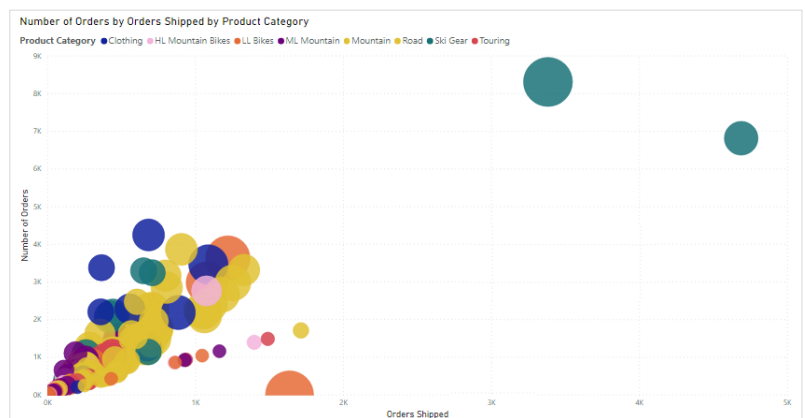
Perform analytics in Power BI

© Copyright Microsoft Corporation. All rights reserved.

109

Identify outliers in data

- Unexpected deviations in data.
- Scatter charts can detect outliers.
- Use DAX to define and analyze outliers dynamically.

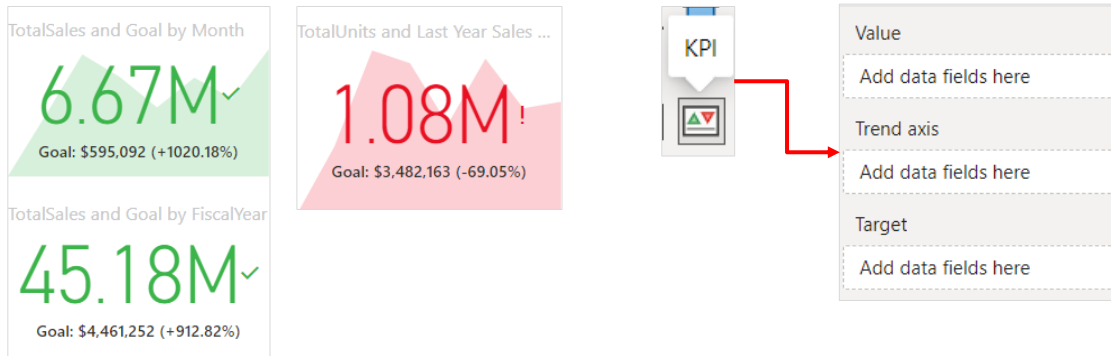


© Copyright Microsoft Corporation. All rights reserved.

110

Key Performance Indicator visual

At-a-glance insights into performance against predefined targets.

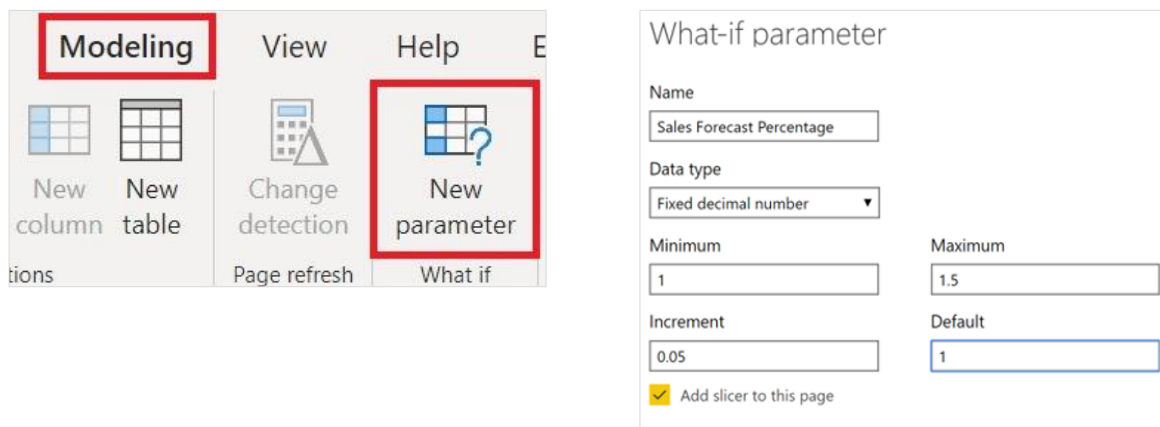


© Copyright Microsoft Corporation. All rights reserved.

111

What-if parameters

Enable scenario planning and interactive exploration for better decision-making.

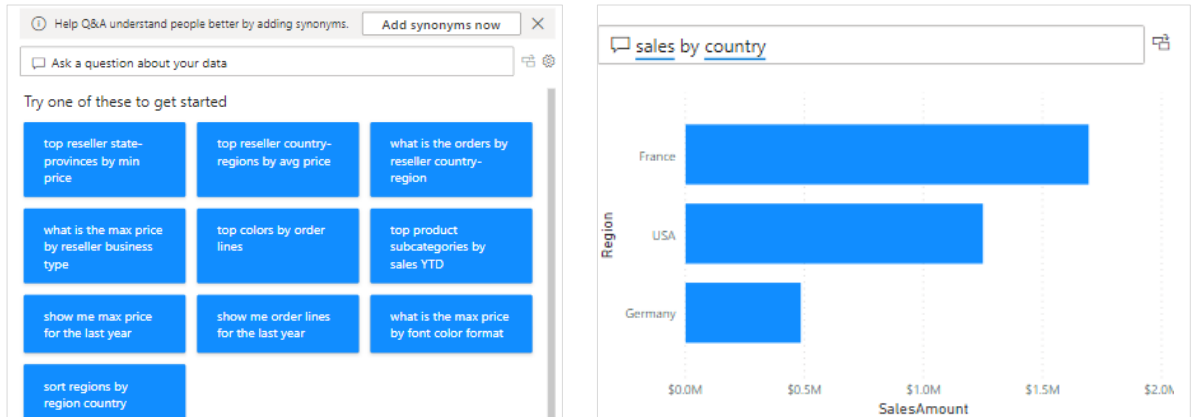


© Copyright Microsoft Corporation. All rights reserved.

112

AI in the Q&A feature

Use natural language to ask questions about data and create visuals for insights.

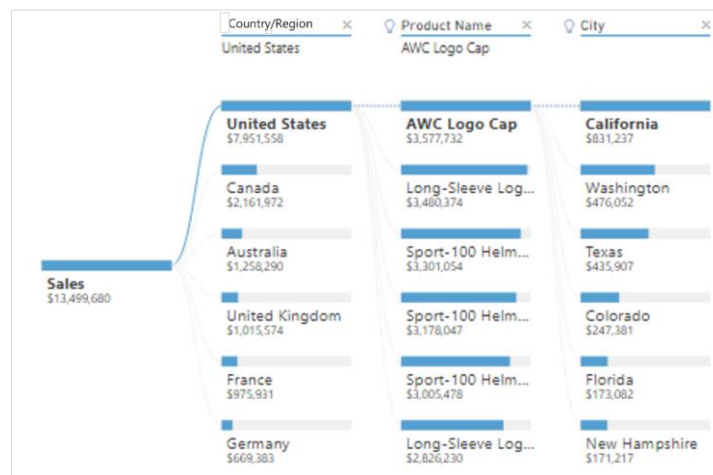


© Copyright Microsoft Corporation. All rights reserved.

113

Decomposition tree visual

Visualize data hierarchies for in-depth exploration and pattern discovery.



© Copyright Microsoft Corporation. All rights reserved.

114

Further reading

Perform analytics in Power BI

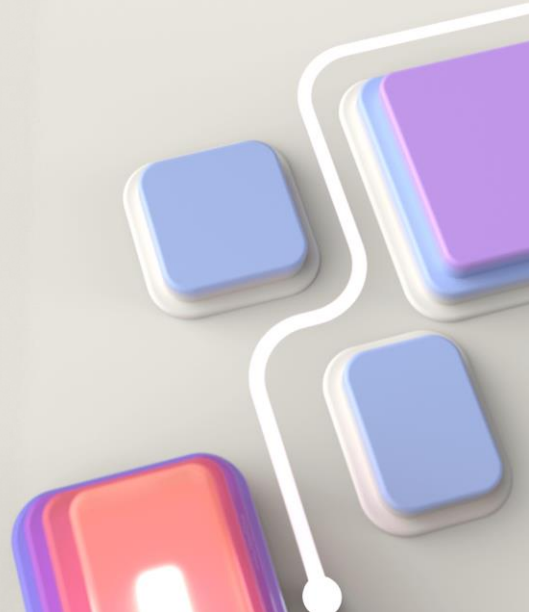
<https://learn.microsoft.com/training/modules/perform-analytics-power-bi/>



© Copyright Microsoft Corporation. All rights reserved.

115

Optimize model performance (optional)



© Copyright Microsoft Corporation. All rights reserved.

116

Use variables to improve performance and readability

Without variable:

Sales YoY Growth =

```
DIVIDE (
    ( [Sales] - CALCULATE ( [Sales], PARALLELPERIOD ( 'Date'[Date], -12, MONTH ) ) ),
    CALCULATE ( [Sales], PARALLELPERIOD ( 'Date'[Date], -12, MONTH ) ) )
```

With variable:

Sales YoY Growth =

```
VAR SalesPriorYear = CALCULATE ( [Sales], PARALLELPERIOD ( 'Date'[Date], -12, MONTH ) )
VAR SalesVariance = DIVIDE ( ( [Sales] - SalesPriorYear ), SalesPriorYear )
RETURN
SalesVariance
```

© Copyright Microsoft Corporation. All rights reserved.

117

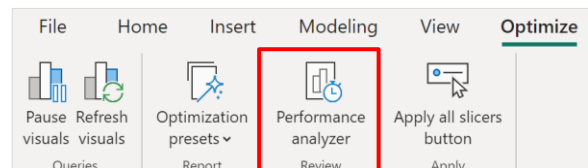
Tune report performance

Performance analyzer in Power BI Desktop

- Record report interactions.
- Review query and rendering times.
- Identify bottlenecks, optimize queries.

© Copyright Microsoft Corporation. All rights reserved.

118



Performance analyzer

Start recording Refresh visuals Stop

Clear Export

Name	Duration (ms)
Recording started (6/19/2023 3:12:20 PM)	-
Cross-highlighted	-
Button	92
Image	91
Sales and Profit Margin by Month	359
DAX query	19
Visual display	28
Other	312
Copy query	
Quantity by Category	302
Slicer	129
Slicer	222

Analyze query plans for optimization opportunities

```
Count Customers =
CALCULATE (
    DISTINCTCOUNT ( Order[ProductID] ),
    FILTER ( Order, Order[OrderQty] >= 5 )
)
```

```
Count Customers =
CALCULATE (
    DISTINCTCOUNT ( Order[ProductID] ),
    KEEPFILTERS (Order[OrderQty] >= 5 )
)
```

Sales by Year		
DAX query		2754
Visual display		57
Other		160
Copy query		

Sales by Year		
DAX query		54
Visual display		57
Other		160
Copy query		

© Copyright Microsoft Corporation. All rights reserved.

119

Further reading

Optimize a model for performance in Power BI

<https://learn.microsoft.com/training/modules/optimize-model-power-bi/>



© Copyright Microsoft Corporation. All rights reserved.

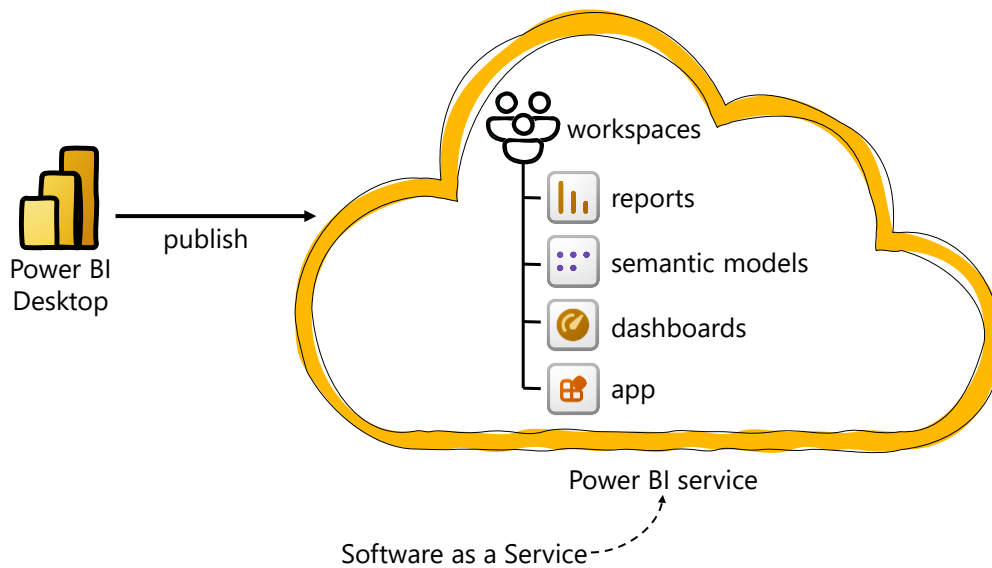
120

Manage workspaces in Power BI

© Copyright Microsoft Corporation. All rights reserved.

121

Understand the Power BI Service

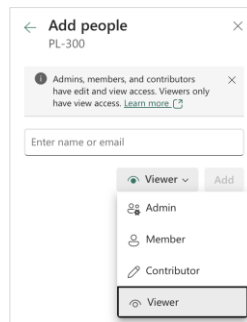


© Cc

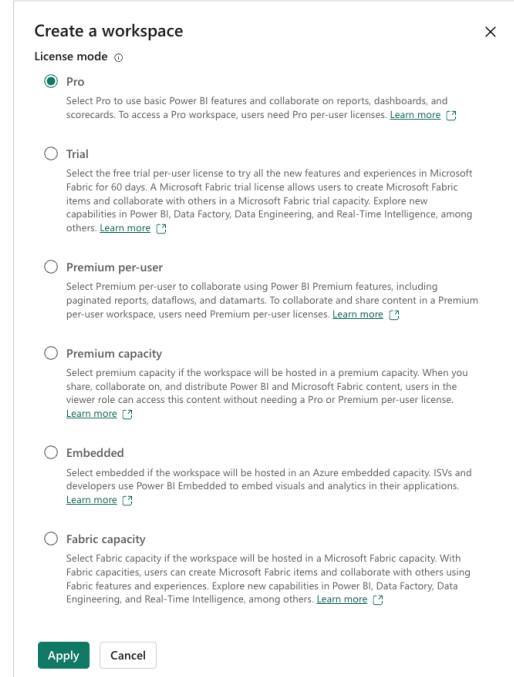
122

Understand Workspaces

- My Workspace vs Shared Workspaces
- License mode
- Grant access to a workspace



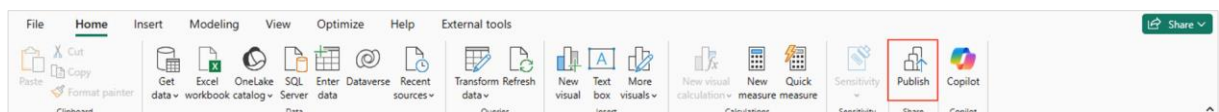
© Copyright Microsoft Corporation. All rights reserved.



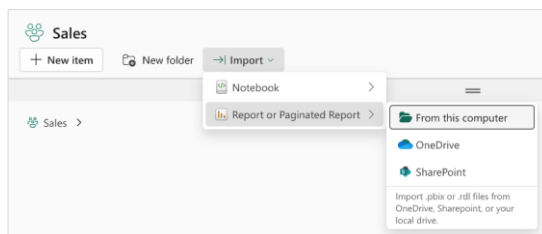
123

Publish to Power BI Service

Publish from Power BI Desktop

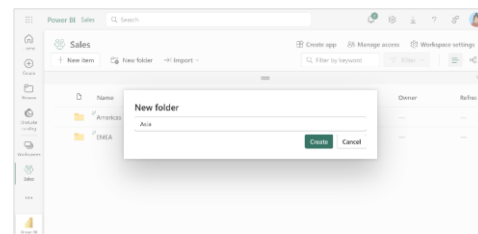


Upload a Power BI report



© Copyright Microsoft Corporation. All rights reserved.

Organize items in folders



124

Further reading

Manage workspaces in Power BI

<https://learn.microsoft.com/training/modules/manage-workspaces-power-bi-service/>



© Copyright Microsoft Corporation. All rights reserved.

125



© Copyright Microsoft Corporation. All rights reserved.

126